

# When True Physical Inputs Help or Hurt a Learned Solubility Model:

## Closure Fidelity and Latent Supervision Govern the Sign of Grounding

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### Abstract

Physics-informed models increasingly route a prediction through a fixed, differentiable physical *closure*—a mechanistic map with no fitted parameters—fed by a learned intermediate. For solid–liquid solubility the closure is COSMO-SAC, which turns a molecule’s surface-charge  $\sigma$ -profile into an activity coefficient; the  $\sigma$ -profile is predicted by a network. The premise is that grounding the closure on truer physical inputs improves it. We show its sign is not fixed. Replacing our model’s learned  $\sigma$ -profiles with the true, externally measured ones (VT-2005) makes prediction *worse* than a free learned head—yet a recent *supervised- $\sigma$*  model reports the opposite. We resolve this by mapping the grounding effect onto two governing axes—how the latent is trained (supervised toward truth vs. shaped end-to-end through the fixed closure) and the closure’s fidelity—anchored by three real systems and a synthetic fidelity dial. On the end-to-end, low-fidelity anchor (ours), grounding hurts because the learned  $\sigma$ -profile is not physical: on a confound-free one-model control it is a systematic, low-rank, out-of-sample-transferable closure-compensating surrogate (converged departure 53%, transfer  $3.3\times$  over identity). An externally-grounded decomposition measures *where* the ceiling sits, and finds it is most likely the closure, not the inputs ( $P \approx 0.78$  under two-way clustering), for the deployed COSMO-SAC-2002 map (our differentiable closure reproduces the NIST reference COSMO-SAC to RMSE 0.003, so this is genuine misspecification, not an implementation artifact). Raising closure fidelity moves the ceiling: a standard modern closure (COSMO-SAC-2010/dsp) removes  $\sim 70\%$  of the closure error and *dissolves the closure-dominance certificate*—the binding term is no longer certifiably the closure. The practical map: if the closure is misspecified and the latent is end-to-end, grounding hurts and the latent is not physical—supervise the latent, or raise closure fidelity. We give the supporting composed-predictor theory (an exact error decomposition, a crystal/activity non-identifiability result, and a physics-tax sign rule); we make no accuracy claim.

## 1 Introduction

Coupling a graph encoder to an explicit thermodynamic bottleneck is a common recipe for out-of-distribution solubility prediction: the model must route its prediction through interpretable physical factors, and each factor can in principle be grounded on abundant single-component data that a black-box predictor cannot exploit. The expectation is that the closer the model’s physical inputs are to the truth, the better it should predict. On the activity branch this expectation fails on our model—yet a recent supervised- $\sigma$  model reports the *opposite*, so the sign of grounding is not a fixed fact. The reason we dwell on our failure is what its *cause* reveals. Having the latent’s true value externally does two things: it lets us *measure* where the ceiling lies and attribute it to

the closure rather than the inputs (the only setting in which that attribution is possible, since a fitted latent has no external oracle), and it lets us hold the learned latent against its truth—which shows that, trained end-to-end through a misspecified closure, the latent is driven off the physical quantity into a systematic, low-rank, out-of-sample-transferable compensating surrogate (confound-free control, §6.3). Both organize into one map: whether grounding helps or hurts is governed by two axes—how the latent is trained (supervised vs. end-to-end) and the closure’s fidelity—which we anchor with three real systems and a synthetic dial, and a standard higher-fidelity closure (COSMO-SAC-2010/dsp) moves the ceiling and dissolves the closure-dominance certificate (§6.2). The composed-predictor framing (§4) and the physics-tax sign rule give the formal account of the sign; the synthetic dial (§6.4) and the pKa closure (§6.6) trace the same two axes. We make no accuracy claim.

**Contributions.** Our thesis is that whether grounding a learned latent in true physics helps or hurts is governed by two axes—how the latent is trained (supervised toward truth vs. shaped end-to-end through the fixed closure) and the closure’s fidelity. We map this, give the mechanism on the corner where grounding hurts, a measurement that locates the ceiling, and the supporting theory and negatives.

- **The grounding paradox (empirical).** Substituting true VT-2005  $\sigma$ -profiles into the differentiable COSMO-SAC closure degrades activity ( $\ln \gamma_2^\infty$ ) and solubility ( $\ln x_2$ ) prediction relative to a free learned activity head, the opposite of the intended benefit (§6.1).
- **A measurement instrument: the closure/insufficiency decomposition.** Because the physical latent  $z^*$  (the true  $\sigma$ -profile) has external ground truth, the true-input excess risk splits *exactly* as  $B = B_{\text{closure}} + B_{\text{insuff}}$  with no model fit (Lemma 3). The decomposition and its estimators are textbook; the new element is exploiting an externally measured latent to bound the split empirically. On the convention-robust evidence—a bound that holds however the COSMO-SAC closure is configured, plus the more-physics-worse effect— $B_{\text{closure}}$  (mis-specified decoder) exceeds  $B_{\text{insuff}}$  (input insufficiency) on the low- $\gamma$  subset we can measure, robustly across the  $k$ -NN, binning, and random-forest insufficiency estimators (only the most conservative ridge does not certify) and, under the law-of-total-variance bound, to leaving out any solvent: there, the binding term is *likely* the closure (§6.2).
- **The mechanism where grounding hurts: a compensating-surrogate latent.** On the end-to-end, low-fidelity anchor the network’s “physical” input does not stay physical: on a confound-free one-model control the predicted  $\sigma$ -profile departs from the true one by 53% in a low-rank (top-two modes 73%), out-of-sample-transferable ( $3.3\times$ ) pattern—a systematic closure-compensating surrogate, not the physical quantity (a  $\sigma$ -grounded model *can* recover the true profile; the objective pulls it away). This is why grounding hurts here, and why a *supervised- $\sigma$*  model finds the opposite (§6.3).
- **The closure-fidelity axis: a modern closure dissolves the closure-dominance certificate.** On the same matched pairs under a reference 3-profile database (COSMO-SAC-2010/dsp, NIST implementation), the standard better-parameterized closures cut the closure error by  $\sim 70\%$  and *dissolve the closure-dominance certificate* ( $P \approx 0.79$  that  $B_{\text{closure}}$  no longer exceeds  $B_{\text{insuff}}$  under two-way clustering); the residual “inputs bind” reading holds unless the true  $B_{\text{insuff}}$  falls below  $\text{MSE}_{2010}/2 \approx 0.38$ , below every  $B_{\text{insuff}}$  estimate we have—so the 2002 closure-dominance is a documented H-bond-kernel inadequacy, and the actionable lesson is to raise closure fidelity, not to enrich the inputs (§6.2).

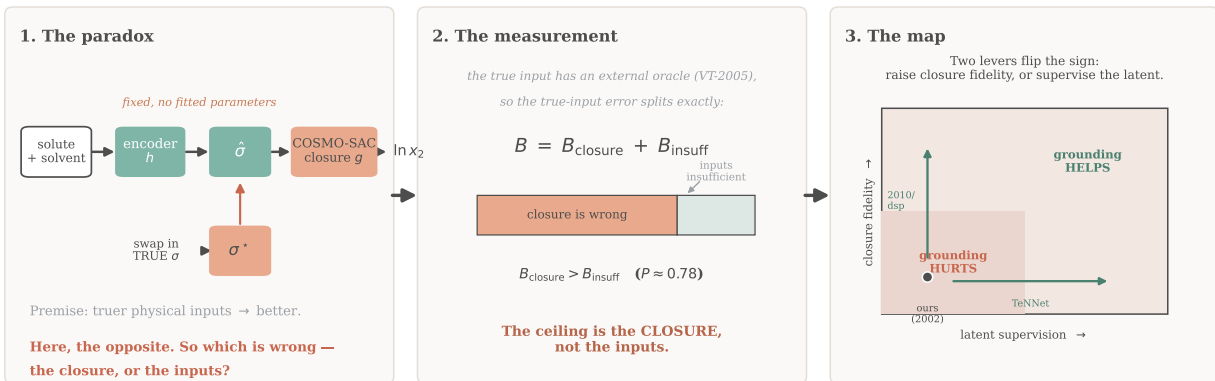


Figure 1: **The paper in one view.** A learned encoder  $h$  predicts a physical latent—a molecular  $\sigma$ -profile—that a fixed COSMO-SAC closure  $g$  (no fitted parameters) turns into a solubility. (1) *The paradox*: replacing the learned  $\hat{\sigma}$  with the *true*, externally measured  $\sigma^*$  (VT-2005) makes the prediction *worse* than a free learned head—so which is wrong, the closure or the inputs? (2) *The measurement*: because the latent has an external oracle, the true-input excess risk splits *exactly* into a closure-misspecification part  $B_{\text{closure}}$  and an input-insufficiency part  $B_{\text{insuff}}$ ; on the matched low- $\gamma$  set the closure binds ( $B_{\text{closure}} > B_{\text{insuff}}$ ,  $P \approx 0.78$ ). (3) *The map*: two levers move the ceiling—raising closure fidelity (COSMO-SAC-2010/dsp) or supervising the latent toward truth (TeNNet-SAC)—so whether grounding helps is governed by two axes, closure fidelity and latent supervision.

- **A negative actionability result (Gate B).** A post-hoc output-space residual placed on the physics prediction recovers calibration but not accuracy; the closure binding cannot be undone downstream (§6.5).
- **A broad, explained negative.** The physics prior does not beat a matched black box on accuracy at any training-data budget—including the low-data regime where an inductive bias should help most (§6.7)—nor, as a trained model, on temperature (though the physical hypothesis *class* does extrapolate there, §6.8); read against the lemmas and the closure measurement, this points to a *structural* ceiling within this closure family rather than an incidental one (§7). We make no accuracy claim.
- **A general account, and honest scoping.** The method is not tied to COSMO-SAC: it is stated for any composed predictor  $\hat{y} = g(h(x))$  with a fixed closure  $g$  over a learned intermediate. We prove three structural lemmas (Lemmas 1–3), a grounding-gap identity (Theorem 1), and a sign rule for the physics tax (Prop. 1). The grounding-gap identity bounds the gap by the closure bias, but only under assumptions that fail here; we therefore read the gap as a *symptom* and let the assumption-light decomposition do the certifying. The sign rule’s critical-budget and aleatoric-floor corollaries predict the two negatives above. The account transfers to synthetic closure families (§6.4) and a second chemical closure (pKa through a Hammett relationship, §6.6, where a trained physics arm does beat its control—though a stratified check shows the gain is not cleanly a closure-fidelity effect); all proofs are elementary (App. B).

## 2 Related work

Two families of methods now dominate solid solubility and solvation prediction, and they make opposite bets about physical structure. Physics-informed pipelines such as SolProp [30] assemble a thermodynamic cycle from separately learned components (solvation free energy, vapor pressure, fusion properties), so that each prediction is decomposable and the intermediate quantities carry physical meaning. Direct models forgo that structure: FastSolv [1] regresses solubility from molecular descriptors with a descriptor-based deep neural network (fastprop), and it is the current leader on new-solute extrapolation, a task on which inter-laboratory disagreement sets a substantial irreducible-error floor. The ordering is unambiguous on accuracy: the black box leads, and the physics-informed cycle trails it. Our work does not try to overturn this ordering. It asks instead where the ceiling of the physics prior comes from, and it uses a matched direct control that shares our encoder so that the accuracy gap is attributable to the closure rather than to representation.

The activity coefficient is the term our closure must supply, and learned surrogates for it have advanced quickly. Winter et al. [34] predict infinite-dilution activity coefficients  $\ln \gamma_2^\infty$  directly from SMILES with a language model, and Sanchez-Medina et al. [26] embed a Gibbs-Helmholtz constraint in a graph network to capture the temperature dependence of  $\ln \gamma_2^\infty$ . These treat  $\ln \gamma_2$  as the regression target. We take the complementary route of retaining a mechanistic activity model, either COSMO-SAC over a predicted  $\sigma$ -profile or NRTL, as a differentiable closure inside the solver, so that the network predicts physical inputs rather than the coefficient itself. That choice is what exposes the failure mode we measure: the misspecification of a fixed mechanistic map can dominate the insufficiency of its learned inputs.

Beyond solubility, the question we pose—when a fixed physical structure helps or hurts a learned model—is the central tension of physics-informed machine learning [11]. Its dominant instantiations impose structure as a differentiable object: physics-informed neural networks enforce a governing equation through the loss [22], while physics-guided and gray-box hybrids compose learned components with fixed mechanistic maps [33]—exactly our  $g(h(x))$ , with  $g$  the mechanistic map. The prevailing expectation is that more physics improves generalization, yet the same structure can degrade it: physics-informed networks have documented failure modes in which the imposed constraint makes optimization or accuracy worse [14]. Our contribution follows from one affordance—an externally measured latent—and does two things: we make that degradation *measurable* (decomposing it into closure misspecification and input insufficiency, with a sign rule for when a fixed-closure prior must lose), and the *same* grounding exposes the learned intermediate, held against its true value, as a closure-compensating surrogate—so the bottleneck’s apparent interpretability is not guaranteed where the closure is misspecified. This compensation is the physics-closure analogue of *concept leakage* in concept-bottleneck models, where a learned concept silently encodes information beyond its nominal meaning and the “interpretation” is misleading [17, 19, 9]; what an external latent oracle adds here is that the leak is *structured*—low-rank and transferable across chemistry, and driven by the closure’s misspecification—rather than diffuse per-example noise. The solubility closure is our worked instance; the pKa closure and synthetic families show the account is not tied to it.

We borrow several diagnostic lenses and claim no new theory from them. The gap between a fixed physical map and reality is the model-discrepancy problem of Kennedy and O’Hagan [12] and its calibration critique [3]; separating whether predictions are right on average from whether they are sharp is the calibration refinement and reliability-resolution decomposition of Murphy and DeGroot [21, 5]; fitting a wrong model still converges to a well-defined pseudo-true limit under the quasi-maximum-likelihood account of White [32]. That the decomposition is only weakly constrained by solubility alone is a matter of parameter sloppiness [28, 8] and of structural versus practical

identifiability [24, 31]. Finally, the risk that apparent skill reflects pipeline choices rather than physics is the concern of underspecification [4], leakage and reproducibility [10], and compositional risk [16]. Our data come from BigSolDB 2.0 [13].

Positioned against this literature, our contribution is not a new model and not an accuracy result. It is a *map* of when grounding a fixed physical closure on an externally measured latent helps or hurts, governed by two axes—closure fidelity and how the latent is trained. The same external ground truth lets us (i) decompose the true-input error into a closure-misspecification part  $B_{\text{closure}}$  and an input-insufficiency part  $B_{\text{insuff}}$  and locate where the ceiling lives (an exactly-proved decomposition identity, plus supporting structural lemmas and a physics-tax sign rule; the location itself is measured, not proved); (ii) show that a standard higher-fidelity closure (COSMO-SAC-2010/dsp) shrinks  $B_{\text{closure}}$  and dissolves the closure-dominance certificate; and (iii) expose the end-to-end learned intermediate, held against its true value, as a low-rank transferable closure-compensating surrogate. Whether feeding true physical inputs helps thus depends on the closure’s fidelity and the latent’s supervision, reconciling our worse-with-true-inputs result with supervised- $\sigma$  models that report the opposite. FastSolv and SolProp remain the reference points for accuracy, which we do not contest.

### 3 Problem setup and notation

For a solute  $s$  in solvent  $v$  at temperature  $T$ , the target is  $y = \ln x_2$ , the log saturated mole fraction. Dissolving a solid trades two costs against each other—breaking the crystal lattice ( $\Phi$ ) and mixing the freed molecules against non-ideal solute–solvent interactions ( $\ln \gamma_2$ )—and solid–liquid equilibrium (SLE) sets the log-solubility to their negative sum:

$$\ln x_2 = - \underbrace{\frac{\Delta H_{\text{fus}}}{R} \left( \frac{1}{T} - \frac{1}{T_m} \right)}_{\Phi(T) \text{ (crystal, depends on } s)} - \underbrace{\ln \gamma_2(x_2, T)}_{\text{activity, depends on } (s, v)}, \quad (1)$$

with the constant- $\Delta C_p$  (heat-capacity-of-fusion) correction omitted for clarity. The activity term is produced by a *differentiable closure*  $g$ : a graph encoder predicts a  $\sigma$ -profile—the distribution of screening (surface) charge density over a molecule, the histogram a COSMO model consumes—for each molecule, and COSMO-SAC (the COnductor-like Screening Model–Segment Activity Coefficient [15]) maps the pair of profiles to  $\ln \gamma_2$ . Write  $z^*$  for the *true* physical latent, the externally measured  $\sigma$ -profile from the VT-2005 database (the Virginia Tech 2005 compilation of quantum-chemically computed screening-charge profiles for  $\sim 1,400$  compounds [20]), and  $g(z^*)$  for the closure output when fed the true profile. The model’s own head instead predicts a *learned* profile  $\hat{z}$  and evaluates  $g(\hat{z})$ ; a matched *DirectGNN* control shares the encoder but emits  $\ln x_2$  with no closure, and is the reference for “what the physics bottleneck costs” (Fig. 2).

**The grounding intervention.** Feeding  $z^*$  in place of  $\hat{z}$  (a  $\sigma$ -oracle) isolates the closure from encoder error: if the closure were well specified,  $g(z^*)$  should be the best available activity estimate. The paradox is that it is not.

**Data.** Solubility measurements are a Bemis–Murcko scaffold split of BigSolDB [13] (grouping molecules by ring-system-plus-linker scaffold, side chains stripped), so test solutes share no scaffold with training; the activity labels  $m$  are experimental infinite-dilution activity coefficients ( $\ln \gamma_2^\infty$ , IDAC) compiled from ThermoML (the NIST/IUPAC archive of experimental thermophysical data [7]), and the true  $\sigma$ -profiles  $z^*$  are the VT-2005 database.

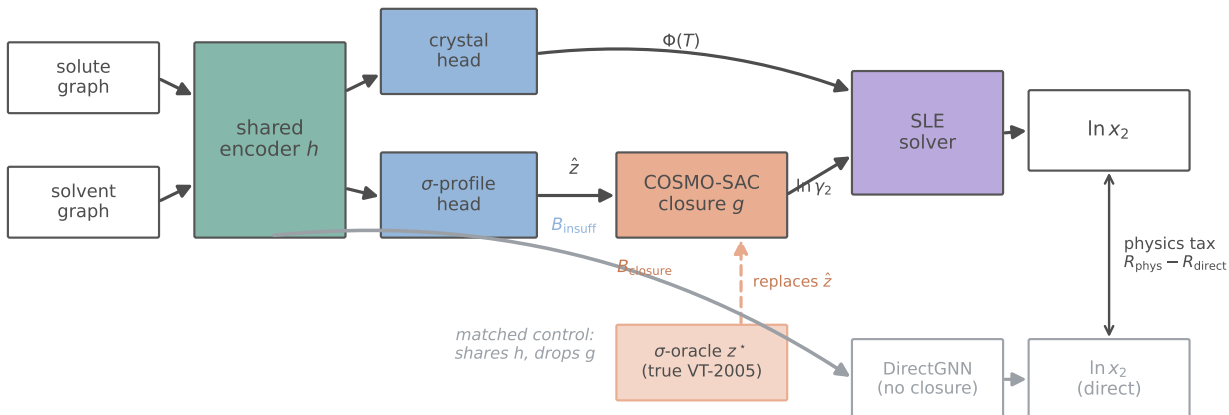


Figure 2: The model and the grounding intervention. Solute and solvent graphs pass through a shared encoder to a crystal head (the ideal term  $\Phi(T)$ ) and a  $\sigma$ -profile head; the predicted profile  $\hat{z}$  feeds the fixed COSMO-SAC closure  $g$ , which returns  $\ln \gamma_2$ , and an SLE solver combines the two into  $\ln x_2$ . Input insufficiency  $B_{\text{insuff}}$  enters at the intermediate  $\hat{z}$  and closure misspecification  $B_{\text{closure}}$  at  $g$ . The  $\sigma$ -oracle replaces  $\hat{z}$  with the true VT-2005 profile  $z^*$  (salmon). DirectGNN (grey) is the matched control—it shares the encoder but drops the closure, so the gap to the physics path is the physics tax  $R_{\text{phys}} - R_{\text{direct}}$ .

## 4 Theory: locating a physics prior’s ceiling in a composed predictor

The results below are stated for a general *composed predictor*  $\hat{y} = g(h(x))$  in which a fixed, non-fitted map  $g$ —the *closure*—consumes a learned physical intermediate  $z = h(x)$ , and are compared against a matched free predictor that shares the encoder but drops  $g$ . Solubility, with  $g$  the COSMO-SAC closure and  $z$  the  $\sigma$ -profile, is one instance; §6.6 gives a second (pKa through a fixed Hammett closure), and §6.4 a controlled synthetic family. Nothing in Lemmas 1–3, Theorem 1, or Proposition 1 is specific to COSMO-SAC. Two quantities organize the section: the closure/insufficiency split  $B = B_{\text{closure}} + B_{\text{insuff}}$ , which is the assumption-light *instrument*, and the observable grounding gap  $\Gamma$ , which is a *symptom* of closure misspecification whose validity we characterize (Fig. 3).

### 4.1 Why a free head can compensate: structural non-identifiability

The crystal/activity split is not identifiable from solubility data alone, which is precisely why an unconstrained head can silently absorb crystal error into the activity term (Fig. 4).

**Lemma 1** (Structural non-identifiability of the SLE split). *Consider the infinite-dilution SLE observable  $\mu(T) = -\Phi(T) - \ln \gamma_2^\infty(T)$  with  $\Delta C_p = 0$ , so that  $\Phi(T) = (\Delta H_{\text{fus}}/R)(1/T - 1/T_m)$  is affine in  $u = 1/T$ , and a symmetric two-suffix activity class ( $\ln \gamma_2 = \ln \gamma_2^\infty (1 - x_2)^2$ ) in which  $\ln \gamma_2^\infty(T)$  is affine in  $u$ . Then, for any design with at least two distinct temperatures,  $\mu$  depends on the parameters only through the two-dimensional statistic (its slope and intercept in  $u$ ), and the*

Table 1: Notation and coined terms used throughout.

| Symbol / term                 | Meaning                                                                    |
|-------------------------------|----------------------------------------------------------------------------|
| $g$ (closure)                 | fixed, non-fitted physical map (COSMO-SAC; a Hammett LFER for pKa)         |
| $h$ (encoder)                 | learned graph network, $x \mapsto z$                                       |
| $z^*, \hat{z}$                | true (oracle) vs. learned physical intermediate (the $\sigma$ -profile)    |
| $m$                           | activity target (experimental $\ln \gamma_2^\infty$ )                      |
| $B_{\text{closure}}$          | closure misspecification, $\mathbb{E}[(\mathbb{E}[m   z^*] - g(z^*))^2]$   |
| $B_{\text{insuff}}$           | input insufficiency, $\mathbb{E}[\text{Var}(m   z^*)]$                     |
| $\Gamma$ (grounding gap)      | $R_{\text{orc}} - R_{\text{e2e}}$ : oracle risk minus best end-to-end risk |
| $\Delta_\infty$ (physics tax) | asymptotic risk of the physics arm minus the matched control               |
| full / res                    | COSMO-SAC with / without the Staverman–Guggenheim combinatorial term       |
| Tier 1 / Tier 2               | convention-independent (load-bearing) / convention-robust corroboration    |

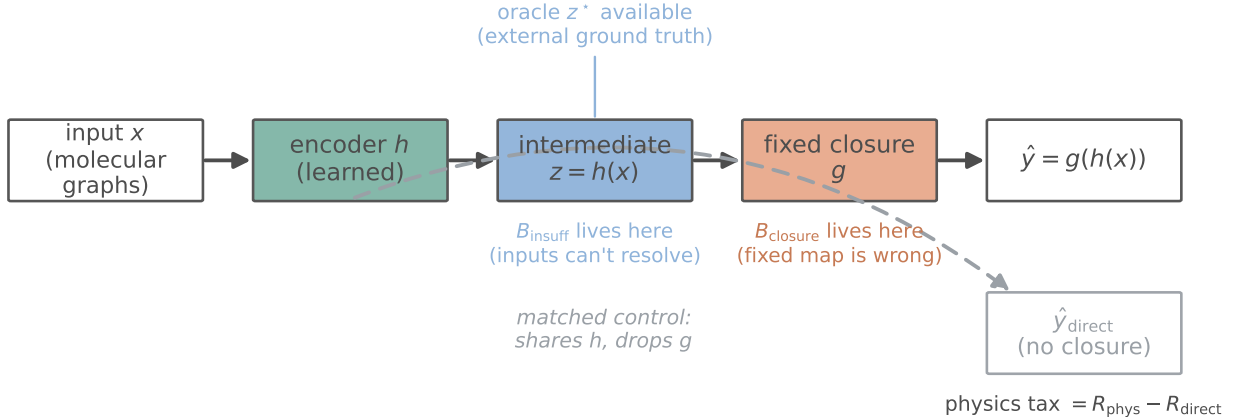
activity intercept and slope already span that plane; hence the crystal parameters  $(\Delta H_{\text{fus}}, T_m)$  are unidentified, the profiled Fisher information for  $\Delta H_{\text{fus}}$  is exactly zero, and the indeterminacy set is two-dimensional. The deficiency is closed only by a rank-completing external constraint (a direct crystal label, or a fixed activity slope). At finite dilution the composition factor  $(1 - x_2)^2$  breaks the exact affinity, leaving a non-zero but near-degenerate information of order  $x_2^2$  (numerically, condition number  $2.3 \times 10^5$ ; §F.1).

This is the standard structural-identifiability picture for a collinear design [24, 31]; we prove it in the SLE form (App. B) and audit it numerically in §F.1. It is *enabling background*, not our headline: it explains why grounding an under-determined factor on external single-component data is necessary, and why compensation between the factors is expected. By itself it says nothing about whether grounding helps. That depends on whether the closure that consumes the grounded inputs is correctly specified, which is the question we make measurable next.

The non-identifiability is not binary but graded by how much of the crystal direction the activity class can absorb.

**Lemma 2** (Class-dependent efficient information). *Let the activity nuisance lie in a class  $\mathcal{H}$  with tangent projection  $\Pi_{\mathcal{H}}$ , and let  $v = \partial\mu/\partial\Delta H_{\text{fus}}$  be the crystal score. The efficient information for  $\Delta H_{\text{fus}}$  is  $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = \sigma^{-2} \|v - \Pi_{\mathcal{H}}v\|^2$ . If  $\mathcal{H}$  contains the crystal score’s span  $\{1, 1/T\}$  (a free intercept and slope) then  $\Pi_{\mathcal{H}}v = v$  and  $\mathcal{I}_{\text{eff}} = 0$ , recovering Lemma 1; if  $\mathcal{H}$  is restricted (the slope is known or constrained) then  $\mathcal{I}_{\text{eff}} > 0$  and the Cramér–Rao lower bound (CRLB) on  $\Delta H_{\text{fus}}$  is finite. (The melting point  $T_m$  is likewise an unknown crystal parameter, but is profiled jointly: its score  $\partial\mu/\partial T_m$  lies in  $\text{span}\{1\} \subset \text{span}\{1, 1/T\}$ , already inside the nuisance tangent whenever the slope is free, so the dichotomy is unaffected by treating  $T_m$  as free.)*

In words: the efficient information is what the data can still pin down about the crystal term *after* the activity model has absorbed everything within its reach. If that model can fit a free intercept and slope in  $1/T$ , it absorbs the entire crystal direction and nothing identifies  $\Delta H_{\text{fus}}$  ( $\mathcal{I}_{\text{eff}} = 0$ ); constraining the slope leaves a little information behind. The proof is the efficient-score construction (App. B): project the crystal score onto the orthogonal complement of the nuisance tangent space; the squared norm of the residual is the efficient information. The numerical audit of §F.1 quantifies both regimes, and shows the honest corollary: merely *restricting* the activity slope leaves  $\text{sd}(\Delta H_{\text{fus}}) \approx 8,251 \text{ J/mol}$  — finite but unusable — whereas *external* crystal labels ( $T_m$  and  $\Delta H_{\text{fus}}$  together), which contribute a score direction outside the activity tangent space, close it to  $\approx 1,689 \text{ J/mol}$ .



the same abstraction, three instances

|                   | input $x$             | encoder $h$ | intermediate $z$  | fixed closure $g$           | target $y$ |
|-------------------|-----------------------|-------------|-------------------|-----------------------------|------------|
| <b>solubility</b> | solute+solvent graphs | MPNN        | $\sigma$ -profile | COSMO-SAC                   | $\ln x_2$  |
| <b>pKa</b>        | molecular graph       | MPNN        | Hammett $\sigma$  | LFER $pK_{a0} - \rho\sigma$ | $pK_a$     |
| <b>synthetic</b>  | features              | MLP         | latent $z$        | deformed teacher            | $m$        |

Figure 3: The general composed predictor  $\hat{y} = g(h(x))$  and its matched control. A learned encoder  $h$  produces a physical intermediate  $z$ ; a *fixed* closure  $g$  maps it to the prediction. Input insufficiency  $B_{\text{insuff}}$  lives in  $z$  (what the intermediate cannot resolve) and closure misspecification  $B_{\text{closure}}$  in  $g$  (where the fixed map is wrong); an external oracle  $z^*$  lets us measure the split. The matched control (grey) shares  $h$  but drops  $g$ , so the gap between them is the “physics tax.” The same abstraction instantiates as solubility, pKa, and the synthetic dial (table).

## 4.2 The closure–insufficiency decomposition

Let  $m$  be the activity target (experimental  $\ln \gamma_2^\infty$ ) and  $g(z^*)$  the closure evaluated on the true latent. The excess risk of the true-input path over the best-possible predictor  $\mathbb{E}[m \mid \cdot]$  (the Bayes predictor) decomposes exactly—into a part the intermediate cannot resolve and a part the fixed map gets wrong (Fig. 5).

**Lemma 3** (Exact closure–insufficiency decomposition and its one-sided bounds). *For a fixed (non-fitted) closure  $g$ ,*

$$\underbrace{\mathbb{E}[(m - g(z^*))^2]}_{\text{MSE of the true-input path}} = \underbrace{\mathbb{E}[\text{Var}(m \mid z^*)]}_{B_{\text{insuff}} \text{ (inputs cannot resolve)}} + \underbrace{\mathbb{E}[(\mathbb{E}[m \mid z^*] - g(z^*))^2]}_{B_{\text{closure}} \text{ (decoder mis-map)}}. \quad (2)$$

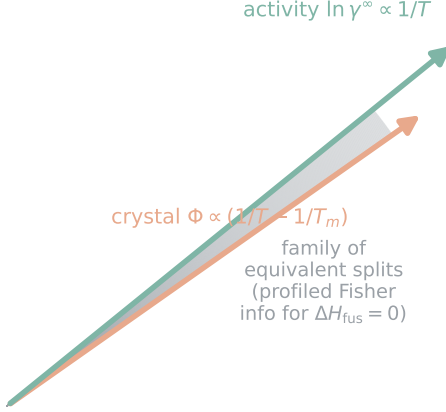
*The cross term vanishes identically because  $\mathbb{E}[m \mid z^*] - g(z^*)$  is  $z^*$ -measurable and  $\mathbb{E}[(m - \mathbb{E}[m \mid z^*]) \mid z^*] = 0$ . The split is exact with no fitting of the closure  $g$ —the cross term vanishes because  $g$  is fixed and  $z^*$  is externally observed—so the total and  $B_{\text{closure}}$ ’s Jensen lower bound are fit-free;  $B_{\text{insuff}} = \mathbb{E}[\text{Var}(m \mid z^*)]$ , however, must be estimated with a conditional-variance smoother whose choice affects the verdict (§6.2).*

Equation (2) is a bias–variance split of model discrepancy in the sense of Kennedy–O’Hagan [12] and Brynjarsdóttir–O’Hagan [3]:  $B_{\text{closure}}$  is the systematic discrepancy of a structurally fixed physical



## Why the crystal/activity split is non-identifiable

(a) solubility alone: split unrecoverable



(b) an off-axis label collapses the band

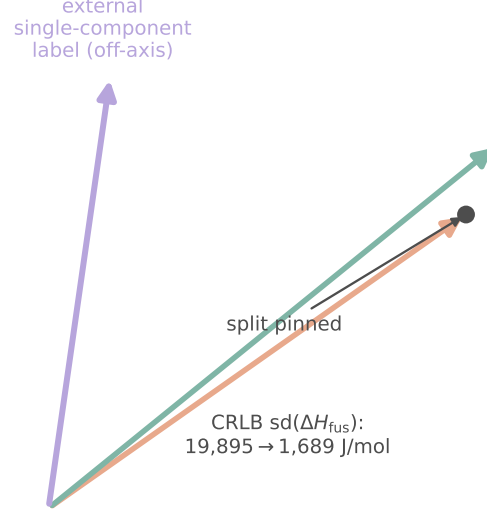


Figure 4: Why the crystal/activity split is non-identifiable from solubility. (a) As functions of  $1/T$  the crystal direction  $\Phi$  and the activity direction  $\ln \gamma^\infty$  are nearly collinear, so a whole band of crystal/activity splits fits the data equally well (the profiled Fisher information for  $\Delta H_{\text{fus}}$  is zero, Lemma 1). (b) An external single-component label points off that axis and collapses the band to a point, pinning the split—the mechanism behind the aux-stream grounding, with the measured Cramér–Rao drop on  $\Delta H_{\text{fus}}$ .

map, and its dominance over  $B_{\text{insuff}}$  is the empirical content of “the ceiling is the closure, not the inputs.” We do not claim (2) as new; the affordance is that the external groundedness of  $z^*$  lets us *measure* the split rather than assume it. This is the move’s distinction from model-discrepancy calibration and auxiliary-data uncertainty quantification, which use held-out physical observations to *fit* a correction to a discrepancy: here the closure is fixed and the latent itself is externally observed, so the map’s error is *attributed* with no correction fitted—at the cost that the affordance runs only where such an external latent oracle exists (fixed-closure predictors with a tabulated or measured latent: LFERs, COSMO / group-contribution closures, thermodynamic cycles), not in gray-box or PINN hybrids whose internal fields have no external ground truth. **The exactness is contingent on  $g$  being fixed:** for a *fitted* decoder the cross term does not vanish and a plug-in point decomposition is invalid (we observe a spurious  $-0.36$  cross term when  $\mathbb{E}[m \mid z^*]$  is replaced by a fitted estimate). We therefore report *one-sided bounds*, not a point split (each proved in App. B):

- $B_{\text{closure}} \geq (\mathbb{E}[m] - \mathbb{E}[g])^2$  by Jensen: a constant offset between the decoder’s mean output and the target is misspecification, not input insufficiency, and this bound needs *no smoothness assumption*. Its one requirement is that  $m$  and  $g$  share the  $\ln \gamma_2^\infty$  reference state, which we check directly (§6.2).
- $B_{\text{insuff}} \leq \mathbb{E}[\text{Var}(m \mid \text{bin}(g(z^*)))]$ , a valid *upper* bound (law of total variance, since a binning of  $g(z^*)$  coarsens  $z^*$ ).

Because  $B_{\text{insuff}} = \mathbb{E}[\text{Var}(m \mid z^*)]$  depends only on  $(m, z^*)$ , it is *independent of the closure conven-*

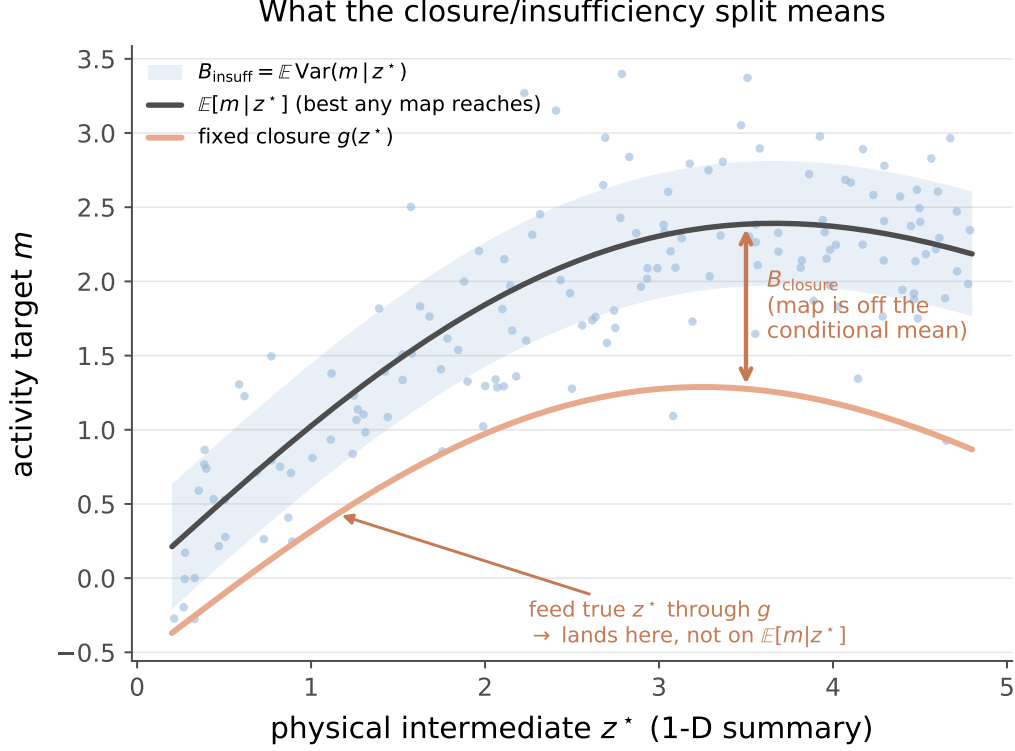


Figure 5: What the closure/insufficiency split means. The activity target  $m$  is scattered against a one-dimensional summary of the physical intermediate  $z^*$ . The vertical spread of the conditional cloud is the input insufficiency  $B_{\text{insuff}} = \mathbb{E} \text{Var}(m | z^*)$ —what no map of  $z^*$  can resolve—while the gap between the conditional mean  $\mathbb{E}[m | z^*]$  (the best any map reaches) and the fixed closure  $g(z^*)$  is the closure misspecification  $B_{\text{closure}}$ . Feeding the true  $z^*$  through  $g$  lands on the lower curve, off the conditional mean; that displacement is the paradox. Schematic; the measured bounds are Fig. 8.

tion: any disagreement between closure variants is decoder disagreement and lands in  $B_{\text{closure}}$ .

### 4.3 The grounding gap: a symptom of closure misspecification

The paradox—true inputs hurting—is a measurable quantity, not a curiosity. Let  $\mathcal{E}$  be the encoder class ( $h : x \mapsto z$ ) and define three population risks of the activity target  $m$ : the oracle risk  $R_{\text{orc}} = \mathbb{E}[(m - g(z^*))^2]$ , the best end-to-end risk  $R_{\text{e2e}} = \inf_{h \in \mathcal{E}} \mathbb{E}[(m - g(h(x)))^2]$ , and the free-predictor (black-box) risk  $R_{\text{free}} = \mathbb{E}[(m - \mathbb{E}[m | x])^2]$ . The *grounding gap* is  $\Gamma := R_{\text{orc}} - R_{\text{e2e}}$ ; the paradox is  $\Gamma > 0$  (a free learned head beats the true input pushed through the closure).

**Theorem 1** (Grounding-gap sandwich). *Assume all risks are finite ( $m, g(z^*) \in L^2$ ;  $g$  and each  $h \in \mathcal{E}$  measurable), (A1) the true encoder is representable,  $\phi^* \in \mathcal{E}$  with  $z^* = \phi^*(x)$ , and (A2) the intermediate is a lossless bottleneck,  $\mathbb{E}[m | x] = \mathbb{E}[m | z^*]$ . Then*

$$0 \leq \Gamma \leq B_{\text{closure}}, \quad \Gamma = B_{\text{closure}} - (R_{\text{e2e}} - R_{\text{free}}), \quad (3)$$

with  $\Gamma = B_{\text{closure}}$  iff  $R_{\text{e2e}} = R_{\text{free}}$  (the composed class realizes the Bayes predictor by distorting the intermediate). In particular a well-specified closure ( $B_{\text{closure}} = 0$ ) gives  $\Gamma = 0$ ; hence under A1–A2,  $\Gamma > 0$  certifies  $B_{\text{closure}} > 0$ .

The proof is elementary (App. B):  $\Gamma \geq 0$  because  $\phi^*$  is a feasible encoder;  $R_{\text{e2e}} \geq R_{\text{free}}$  because every  $g(h(x))$  is  $\sigma(x)$ -measurable; and  $R_{\text{free}} = B_{\text{insuff}}$  under A2. Thus  $\Gamma$  is exactly the closure bias the encoder can *absorb* by feeding  $g$  a wrong-but-useful intermediate—the formal content of Lemma 1’s “a free head silently compensates.”

**$\Gamma$  is a symptom, not a certificate.** Both assumptions are load-bearing and neither holds cleanly here. A1 is *empirically doubtful*: the encoder is transfer-limited (a linear probe onto descriptors drops from train  $R^2 \approx 0.76$  to test  $R^2 \approx 0.45$  out of distribution, a proxy for—not a direct measure of—representability of  $z^*$ ), making  $\phi^* \in \mathcal{E}$  unlikely across scaffolds, so effectively  $\phi^* \notin \mathcal{E}$  and even  $\Gamma \geq 0$  is not guaranteed. A2 is unverifiable without the oracle and physically dubious (the  $\sigma$ -profile is a lossy summary of the graph). When A2 fails,  $\Gamma > 0$  *conflates* two causes: closure misspecification ( $B_{\text{closure}} > 0$ ) and input insufficiency (a correct closure driven off-support to recover information the intermediate discarded). A controlled construction with a *well-specified* closure and a lossy intermediate gives  $B_{\text{closure}} = 0$  yet  $\Gamma > 0$  (§6.4, and the pKa ortho sweep of §6.6). Certification of closure-dominance therefore rests on the convention-independent *decomposition* (Lemma 3), which needs neither A1 nor A2: in solubility the Jensen bound and the insufficiency estimates place  $B_{\text{closure}}$  above  $B_{\text{insuff}}$  *directly* (§6.2), and the grounding gap only corroborates it. This is the paper’s method: *the decomposition is the instrument,  $\Gamma$  is the cheap symptom that motivates it, and separating its two causes is what an externally measured latent buys.*

#### 4.4 The physics tax and the critical data budget

The gap is a population statement; the deployed question is whether the *trained* physics model beats the *trained* black box at sample size  $n$ . Writing each expected risk as an asymptote plus estimation variance,  $\mathbb{E}[\hat{R}_{\text{phys}}(n)] \approx R_{\text{e2e}} + V_{\text{phys}}(n)$  and  $\mathbb{E}[\hat{R}_{\text{direct}}(n)] \approx R_{\text{direct},\infty} + V_{\text{direct}}(n)$  with  $V(n) \downarrow 0$ , the *physics tax* is

$$\mathcal{T}(n) \approx \Delta_{\infty} + [V_{\text{phys}}(n) - V_{\text{direct}}(n)], \quad \Delta_{\infty} := R_{\text{e2e}} - R_{\text{direct},\infty}. \quad (4)$$

**Proposition 1** (Sign of the physics tax). (i)  $\mathcal{T}(n) < 0$  (*physics helps*) iff  $V_{\text{direct}}(n) - V_{\text{phys}}(n) > \Delta_{\infty}$ : the variance a constrained prior saves must exceed its asymptotic bias. (ii) If  $\Delta_{\infty} > 0$  and the variance gap vanishes ( $V_{\text{direct}} - V_{\text{phys}} \rightarrow 0$ ), there is a critical budget  $n^*$  beyond which  $\mathcal{T}(n) > 0$  for all  $n > n^*$ . (iii) Since  $V_{\text{direct}} - V_{\text{phys}} \leq V_{\text{direct}}$ , as the black box nears the aleatoric floor ( $V_{\text{direct}} \rightarrow 0$ ) the tax tends to  $\Delta_{\infty}$ ; this limit is  $\geq 0$  when the control is asymptotically Bayes ( $R_{\text{direct},\infty} = R_{\text{free}}$ ), not merely from  $V_{\text{direct}} \rightarrow 0$ .

When the control is asymptotically Bayes and A2 holds,  $\Delta_{\infty} = B_{\text{closure}} - \Gamma$ , tying the asymptotic tax to the *un-compensatable* part of the closure bias; in general  $\Delta_{\infty}$  is read off the large- $n$  asymptote. Both corollaries are *loosely consistent* with the data—the data-efficiency null (a positive tax at every budget, §6.7, single-seed) and the novelty inversion (the tax heaviest on the *best-covered* solutes, §F.3, exploratory) point in directions the proposition allows—but neither is a controlled test: in the data-efficiency sweep the physics arm is far more under-converged than its control at the shared light budget (its  $R^2$  falls to  $-0.79$  where the control holds near  $+0.22$ ), so the within-budget gap conflates inductive bias with optimization difficulty and we do not map it to a specific case. We prove the structural parts (existence of  $n^*$  given  $\Delta_{\infty} > 0$  and a vanishing variance gap; the floor inequality); the shapes of  $V(n)$  are empirical.

## 4.5 Supporting structural facts

Three further consequences of the same collinear geometry are used informally and are not stated as theorems. (i) *Temperature extrapolation*. Within the affine-in- $1/T$  activity family, same-pair prediction at a temperature  $T^*$  outside the training range is an affine extrapolation whose error is controlled by the fitted-slope variance times  $|1/T^* - 1/\overline{T}|$ , whereas an unstructured temperature encoder is unconstrained beyond its support. This is the one place the physical *structure* is expected to help; §6.8 reports the class-level margin, which is a property of the hypothesis class and not an accuracy claim for the trained model (indeed the trained models trail the van’t Hoff baseline). (ii) *Geometry of compensation*. As the training window  $\Delta T \rightarrow 0$ ,  $1/T$  becomes nearly constant, the crystal and activity design columns become collinear, and the smallest Fisher eigenvalue tends to zero (the  $\Delta T$  sweep of §F.1 reaches conditioning  $10^{16}$  at  $\Delta T = 10$  K), so an optimizer prefers to split error between the factors rather than fit either correctly. (iii) *Grounding design*. Each single-component stream rank-completes a different deficient direction of Lemma 1; this is the design rule behind the auxiliary streams, though (as §F.2 shows for a dose-limited crystal run) it does not by itself promise an end-task gain.

## 4.6 What we prove, and what we do not claim

We prove the three structural lemmas the measurement rests on — the split’s non-identifiability (Lemma 1), its class-dependence (Lemma 2), and the exact closure/insufficiency decomposition with its one-sided bounds (Lemma 3) — together with the grounding-gap sandwich (Theorem 1) and the sign rule for the physics tax (Prop. 1); all proofs are elementary (App. B). We are deliberate about their reach. Theorem 1 certifies closure misspecification *only* under A1–A2, and both fail or are unverifiable in our setting (A1 empirically, A2 in principle); we therefore rest closure-dominance on the assumption-light decomposition and use  $\Gamma$  as a corroborating symptom, not a certificate. Proposition 1 predicts the *sign* of the tax from measurable quantities, not an optimum. In particular there is no “optimal grounding” theorem and no necessary-and-sufficient condition for the decomposition to improve out-of-distribution accuracy — the honest empirical answer (§6) is that it does not. The lemmas are read against, not in competition with, known results on model discrepancy [12, 3], parameter sloppiness [28, 8], structural and practical identifiability [24, 31], estimation under misspecification [32], and underspecification [4]; the calibration-versus-accuracy separation of §6.5 is the classical refinement decomposition of a probability forecast [21, 5].

# 5 Methods

The model is a controlled comparison: a shared graph backbone feeds either an explicit thermodynamic bottleneck (*TGNNSol*) or a direct regressor (*DirectGNN*) that shares the backbone but emits  $\ln x_2$  with no physics. The gap between them measures what the bottleneck costs or buys. This section describes the shared backbone, the three interchangeable activity closures, the solid–liquid-equilibrium (SLE) solver and its implicit differentiation, the grounding intervention, and the training protocol. Exact hyperparameters, the full feature list, and the complete loss taxonomy are in the SI (§A).

## 5.1 Data, splits, and labels

Solubility measurements are drawn from BigSolDB 2.0 [13] and partitioned by Bemis–Murcko scaffold (`scripts/prepare_data.py`) so that every test solute is unseen at training; row membership is

set by a melting-point-independent miscibility/structure filter, and crystal  $T_m$ ,  $\Delta H_{\text{fus}}$ , Hansen parameters, and infinite-dilution activity coefficients  $\ln \gamma_2^\infty$  are merged on as per-solute labels where available. Two data hazards are handled explicitly. Melting-point tables arrive in either °C or Kelvin and are auto-detected by column median (a past double-conversion inflated every  $T_m$  by 273 K, which propagates directly into the ideal term); and the seeded scaffold split is not stable across pipeline versions, so every regeneration is treated as a new split with all downstream artifacts recomputed. Crystal supervision is scarce: the test and validation splits contain essentially no measured  $\Delta H_{\text{fus}}$ , which is the quantitative backbone of the identifiability argument (§4). External single-component data used for grounding are an open crystal-property pool ( $\sim 15,000$  melting points), the VT-2005  $\sigma$ -profile database, and ThermoML infinite-dilution activity coefficients; each is filtered to exclude any molecule whose scaffold appears in the validation or test split.

## 5.2 Graph encoder and pair representation

Solute and solvent are encoded by a *shared* graph network with three interchangeable backbones: a message-passing network (MPNN), a hybrid local-message/global-attention graph transformer (GPS [23]), and a physics-channelled variant, Thermodynamically-Informed Message Passing (TIMP), that splits messages into dispersive, polar, and hydrogen-bonding channels supervised toward the Hansen solubility parameters (the dispersive, polar, and hydrogen-bonding components of a molecule’s cohesive energy density) to prevent channel collapse.<sup>1</sup> Small hydride solvents such as water are given explicit hydrogens so the polar and hydrogen-bond channels activate. Solute and solvent embeddings interact through a bipartite cross-attention/message-passing block and a physics-aware readout, and are combined into a pair representation  $[g_s, g_v, g_s \odot g_v, |g_s - g_v|]$  passed through an MLP, optionally augmented with RDKit descriptors, Morgan fingerprints, and an ionic-context feature. All heads below consume this pair representation; the DirectGNN control consumes the identical representation.

## 5.3 Crystal head and the ideal term

A fusion head predicts the two crystal properties that define the ideal (activity-free) solubility, with physically bounded parametrisations  $T_m = 100 + 600 \sigma(\cdot) \in [100, 700]$  K and  $\Delta H_{\text{fus}} = S_H \text{softplus}(\cdot) > 0$ ; an optional entropy-coupled mode instead predicts a Walden-constrained fusion entropy (Walden’s rule: rigid organics have a near-constant entropy of fusion,  $\sim 57 \text{ J mol}^{-1} \text{ K}^{-1}$ ) and sets  $T_m = \Delta H_{\text{fus}} / \Delta S$ . When a group-contribution  $T_m$  prior is used it is affine-calibrated on the training set before entering the head. The crystal properties give the ideal term

$$\Phi(T) = \frac{\Delta H_{\text{fus}}}{R} \left( \frac{1}{T} - \frac{1}{T_m} \right), \quad (5)$$

the constant- $\Delta C_p$  correction omitted (its magnitude is audited in the SI, §E).

## 5.4 Three interchangeable activity closures

The activity term  $\ln \gamma_2$  is produced by one of three closures, selected by configuration, all differentiable and all consuming the same pair representation.

<sup>1</sup>TIMP is a physics-motivated backbone we introduced to test whether a thermodynamically structured encoder helps; across our runs it does not beat the plain MPNN, consistent with the encoder being transfer-limited rather than expressivity-limited (a linear probe reaches train  $R^2 \approx 0.76$  but test  $R^2 \approx 0.45$ ).

**NRTL (Non-Random Two-Liquid [25]).** A head predicts the two interaction energies and a non-randomness factor,  $\tau_{12}^{\text{ref}}, \beta_{12}, \tau_{21}^{\text{ref}}, \beta_{21}, \alpha \in [0.1, 0.6]$ , with a temperature law  $\tau_{ij}(T) = \tau_{ij}^{\text{ref}} + \beta_{ij}(1/T - 1/T_{\text{ref}})$ ; an optional UNIFAC (UNQUAC Functional-group Activity Coefficients [6]) group-contribution prior regularises the  $\tau$ 's. This is the default and the most expressive activity model.

**Single-parameter (`gamma_inf`).** Collapsing the NRTL closure to a symmetric one-parameter form yields the analytic infinite-dilution law  $\ln \gamma_2(x_2, T) = \ln \gamma_2^\infty(T) (1 - x_2)^2$ , with  $\ln \gamma_2^\infty$  affine in  $1/T$ . This is the minimal activity model and the setting in which the identifiability analysis is cleanest (§4).

**Differentiable COSMO-SAC.** A  $\sigma$ -profile head emits, for each molecule, a 51-bin screening-charge-density profile  $p(\sigma)$  (area per bin, summing to the cavity area  $A$ ). A parameter-free COSMO-SAC layer (Lin–Sandler / VT-2005 conventions) maps the pair of profiles to  $\ln \gamma_2$  by a damped fixed point for the segment activity coefficient  $\Gamma_S$  (distinct from the grounding gap  $\Gamma$ ),

$$\Gamma_S(\sigma_m) = \left[ \sum_n p_{\text{norm}}(\sigma_n) \Gamma_S(\sigma_n) \exp(-\Delta W(\sigma_m, \sigma_n)/RT) \right]^{-1}, \quad (6)$$

with a precomputed exchange-energy matrix  $\Delta W$  (electrostatic misfit plus a hydrogen-bonding term); the restoring contribution is  $\ln \gamma_2^{\text{res}} = (A_2/a_{\text{eff}}) \sum_m p_2^{\text{pure}}(\sigma_m) [\ln \Gamma_{S,\text{mix}}(\sigma_m) - \ln \Gamma_{S,2}^{\text{pure}}(\sigma_m)]$ , and an optional Staverman–Guggenheim combinatorial term uses the molecular cavity volume and area. The combinatorial size factor  $r_i = V_i/r_0$  takes the *true* COSMO cavity volume ( $V_{\text{cosmo}}$ ) so that it shares the cavity basis of the area  $A_i$ . This closure — COSMO-SAC evaluated over a network- *predicted*  $\sigma$ -profile, and the intervention that swaps that profile for the measured one — is the object of study in the grounding analysis (§6.1).

## 5.5 The SLE solver and implicit differentiation

Given  $\Phi(T)$  and a closure for  $\ln \gamma_2$ , the saturated mole fraction solves  $\ln x_2 = -\Phi(T) - \ln \gamma_2(x_2, T)$ , a fixed point in  $x_2$  because  $\ln \gamma_2$  depends on composition. We iterate a damped map

$$x_2^{(k+1)} = \lambda \exp(-\Phi(T) - \ln \gamma_2(x_2^{(k)}, T)) + (1 - \lambda) x_2^{(k)}, \quad (7)$$

with closure-specific kernels for the NRTL, `gamma_inf`, and COSMO-SAC paths. Rather than back-propagating through the iterations, gradients use the implicit-function theorem at the converged  $x_2^*$ ,

$$\frac{\partial \ln x_2^*}{\partial \theta} = - \frac{\partial(\Phi + \ln \gamma_2)/\partial \theta}{1 + x_2^* \eta}, \quad \eta = \frac{\partial \ln \gamma_2}{\partial x_2}, \quad (8)$$

which is exact at the fixed point, memory-constant in the iteration count, and exposes an explicit stability criterion: the gradient is well conditioned unless  $1 + x_2^* \eta \rightarrow 0$ . Iteration counts (train 16, eval 30) were set by a convergence audit — a shorter  $n=8$  schedule failed a  $5.86\text{-}\ln x_2$  convergence test on strongly non-ideal pairs (SI §A).

## 5.6 Adaptive correction and the grounding intervention

A bounded, gated “adaptive physics correction” perturbs the physical parameters ( $T_m, \Delta H_{\text{fus}}, \tau$ ) rather than the output, re-solves (7), and blends  $\ln x_2^{\text{final}} = \ln x_2^{\text{phys}} + (1 - w) \text{clip}(\Delta)$ ; keeping the correction in parameter space preserves the physical decomposition. The *grounding intervention*

(a  $\sigma$ -oracle) replaces the predicted  $\sigma$ -profile  $\hat{z}$  with the true VT-2005 profile  $z^*$  before the closure, matched by canonical SMILES. It is applied three ways to rule out wiring artifacts: at evaluation only, injected at training time so the rest of the model co-adapts, and as a channel-swap control; an analogous override substitutes true crystal targets. The activity target used for the closure/insufficiency decomposition (§6.2) is evaluated at infinite dilution,  $\ln \gamma_2^\infty = \lim_{x_2 \rightarrow 0} \ln \gamma_2$ , at 298 K.

### 5.7 Training: three-phase curriculum and single-component grounding

Optimisation follows a fixed three-phase curriculum: (P1) auxiliary-property pretraining on crystal, Hansen, and  $\ln \gamma_2^\infty$  targets with no SLE loss; (P2) full SLE training, with the adaptive correction unfrozen partway and an optional crystal→pair weight ramp; (P3) low-learning-rate stabilisation with monotonicity and van’t Hoff regularisers. External single-component data enter through *grounding streams*: each stream is a sidecar of *self-solvent* rows (solvent = solute, no solubility label, only the target factor’s mask active), interleaved into P1/P2 at a configurable step budget, with a mandatory scaffold-leak guard excluding any pool molecule whose scaffold is in validation or test. The reference streams ground the crystal factor (melting-point pool) and the activity factor ( $\sigma$ -profile database); this single-component-grounding pattern is the mechanism by which an otherwise non-identifiable factor (§4) is pinned by abundant external data a black-box predictor cannot use.

### 5.8 Objective

The training loss is a weighted sum with per-phase weights. Only a small set of terms is load-bearing for the claims in this paper: a bin-weighted Huber on  $\ln x_2$ ; auxiliary losses on the  $\sigma$ -profile (an earth-mover distance), on crystal  $T_m/\Delta H_{\text{fus}}$ , on  $\ln \gamma_2^\infty$ , and on Hansen parameters; and structural regularisers for temperature monotonicity and the van’t Hoff slope. The remaining terms (channel orthogonality, contrastive, mixture-of-experts balance, and various priors) are auxiliary; the complete taxonomy and weights are tabulated in the SI (§A).

### 5.9 The DirectGNN control

DirectGNN shares the encoder, interaction, readout, and pair representation, adds a thermometer temperature encoding, and maps directly to  $\ln x_2$  with a bin-weighted Huber loss and no heads, closure, or solver. Because it shares everything but the thermodynamic bottleneck, the TGNNSolv–DirectGNN gap is attributable to the bottleneck rather than to representation, and DirectGNN is the reference for every “physics tax” comparison in §6.

### 5.10 Reproducibility

All controlled comparisons are run at three seeds and reported as mean±sd on a cross-arm intersection lock (rows supervised and finite in every arm). Solver iteration counts, COSMO-SAC constants, the exact feature list, per-phase loss weights, and the compute ledger (which results are full-corpus GPU runs versus CPU-reproducible proxies) are given in the SI (§A, §H); the named analysis scripts regenerate every figure and table from committed artifacts.

Table 2: What the paper establishes, graded by strength of evidence. *Certified* claims hold under any closure configuration or are exactly proved; *likely* claims are measured but sub-threshold or one-sidedly bounded on the matched  $n=60/n=44$  sets; *exploratory* claims are directional or conjectural. We make no accuracy claim.

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Certified</b>   | <p>The grounding paradox: true VT-2005 <math>\sigma</math>-profiles make the model <i>worse</i> than a free learned head (3 seeds, <math>n=5608</math>, surviving three injection controls).</p> <p>The exact split <math>B = B_{\text{closure}} + B_{\text{insuff}}</math> for a fixed closure (Lemma 3).</p> <p>The differentiable closure reproduces the NIST reference COSMO-SAC-2002 to RMSE <math>0.003 \ln \gamma</math>, so <math>B_{\text{closure}}</math> is genuine misspecification, not an implementation artifact.</p>                                                                                                                                                                                                       |
| <b>Likely</b>      | <p>On the low-<math>\gamma</math> <math>n=60</math> set the closure binds: <math>B_{\text{closure}} \geq 0.85 &gt; 0.62</math>, <math>P \approx 0.78</math> (two-way clustering; the margin CI spans zero; not certified at conventional confidence).</p> <p>A standard modern closure (COSMO-SAC-2010/dsp) dissolves that certificate (<math>P \approx 0.79</math>); the reading turns to the inputs for any true <math>B_{\text{insuff}} &gt; \text{MSE}_{2010}/2 \approx 0.38</math>, below every <math>B_{\text{insuff}}</math> estimate.</p> <p>The end-to-end latent is a low-rank (top-two 73%), out-of-sample-transferable (<math>3.3\times</math>) closure-compensating surrogate (<math>n=44</math>, one converged control).</p> |
| <b>Exploratory</b> | <p>Transfer of the closure-dominance to the finite-composition/high-<math>\gamma</math> regime (supported by in-regime signs, not measured there).</p> <p>The <math>\gamma</math>-stratified strengthening of <math>B_{\text{closure}}</math> (thin, estimator-dependent).</p> <p>The pKa/Hammett second closure as a fidelity-axis anchor (a cautionary complement, not a clean one).</p>                                                                                                                                                                                                                                                                                                                                                 |

## 6 Results

Results build the map. The paradox (§6.1) motivates; the decomposition then *measures* where the ceiling sits and a standard higher-fidelity closure moves it (the closure-fidelity axis, §6.2); the compensating-surrogate mechanism (§6.3) explains the end-to-end/low-fidelity corner where grounding hurts. The remaining sections—the synthetic dial (§6.4), the output-residual control (§6.5), the pKa closure (§6.6), and the data-efficiency and chemistry maps—stress-test and generalize the account.

### 6.1 The grounding paradox

Feeding true  $\sigma$ -profiles into the closure degrades it rather than improving it, in both the activity and the solubility target, and it fails the way a *misspecified map* does—not the way missing information would. In the solubility target the effect erases any positive  $R^2$  (Fig. 6): the model predicts solubility better *from its own learned  $\sigma$ -profile* than when *given the true measured one* (MAE 1.85 vs 2.25, in  $\ln x_2$  units). Two facts pin the cause on the map. First, restoring the physically *more* complete Staverman–Guggenheim combinatorial term (the entropic contribution from solute–solvent size and shape mismatch) *increases* error—what a wrong map does, not what a missing input would. Second, on the activity target ( $\ln \gamma_2^\infty$ ,  $n=60$  matched pairs) the fixed true-input closure sits about  $4\times$  above the irreducible-variance floor a flexible regressor reaches (§6.2), under-predicting  $\ln \gamma_2^\infty$  with a systematic bias (Fig. 7). The paradox survives three controls—the evaluation-time oracle, injection of the true profile *at training* (MAE 1.98), and a channel-swap (2.08), all worse than the learned- $\sigma$  arm—ruling out a wiring artifact. The  $\sigma$ -oracle per-arm values are in §C; the true-at-train and channel-swap figures are the inline values above.



The grounding paradox: true  $\sigma$ -profiles give the worst arm (3 seeds,  $n = 5608$ )

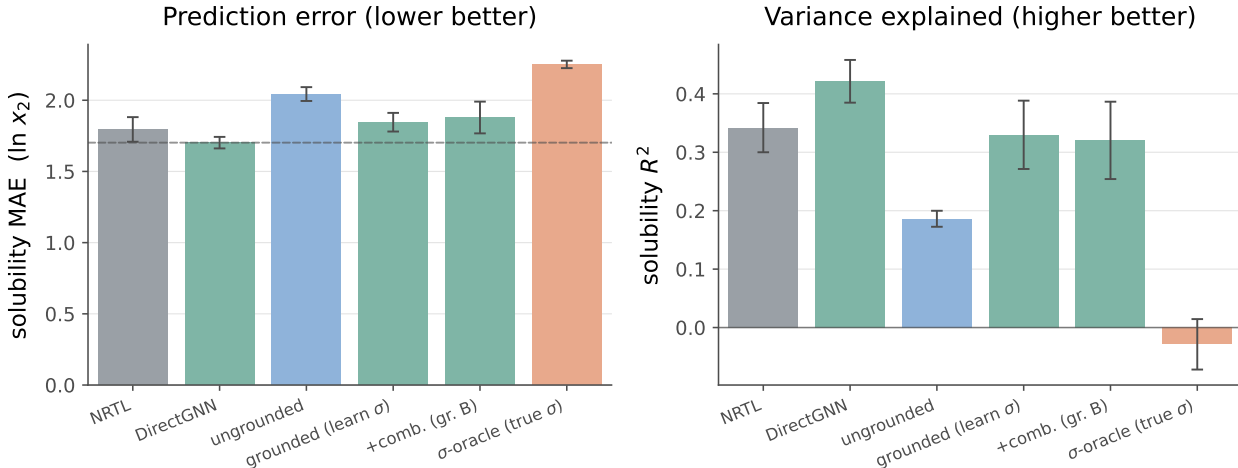


Figure 6: Controlled comparison on the scaffold split (3 seeds, mean $\pm$ sd,  $n=5608$ ; metrics are the 3-seed mean of the cross-arm intersection lock — rows supervised and finite in every arm — from `run_e5_comparison.py`). DirectGNN shares the encoder but has no thermodynamic closure; the physics closures trail it (the “physics tax”). Replacing the model’s learned  $\sigma$ -profile with the *true* VT-2005 profile (the  $\sigma$ -oracle, salmon) gives the *worst* arm, collapsing  $R^2$  to  $\approx 0$  ( $-0.03 \pm 0.04$ ): by MAE the model predicts solubility distinctly better from its invented  $\sigma$  (1.85) than from the measured one (2.25). Per-arm exact values are transcribed in the SI.

## 6.2 Measuring the ceiling: the closure dominates

Because the true latent is externally observed, we can locate the prior’s ceiling and attribute it to the map rather than the inputs—the measurement instrument at the core of the map. We apply the decomposition of §4.2 to the  $n=60$  matchable activity pairs (298 K, infinite dilution), with  $z^*$  the true concatenated solute/solvent  $\sigma$ -profiles and  $m$  the experimental  $\ln \gamma_2^\infty$ ; Table 3 reports the bounds. The set spans 41 solutes and 17 solvents in 102 profile dimensions, with two formamide pairs dropped on a canonical-SMILES tautomer mismatch.

We grade the evidence in two tiers: *Tier 1* claims hold for *any* COSMO-SAC configuration and are the load-bearing ones, while *Tier 2* claims are configuration-robust corroboration that is not strictly independent of the model fit.

**The load-bearing separation (leakage-immune, Tier 1).** We subtract the leakage-immune law-of-total-variance  $B_{\text{insuff}}$  upper bound—the most conservative of our estimators, since the  $z^*$ -neighbour estimators ( $k$ -NN, RF) are deflated by near-duplicate leakage and so *inflate*  $B_{\text{closure}}$ —from the total error. On the model’s *deployed* residual-only convention this gives  $B_{\text{closure}} \gtrsim 0.85$ , well above  $B_{\text{insuff}} \lesssim 0.62$  (Table 3). The bound conditions on  $g(z^*)$ , so we report it per convention; it is the claim we lean on, and it needs no smoothness assumption beyond the validity of the  $B_{\text{insuff}}$  upper bound. We stress-test the separation directly (App. C), and doing so exposes an identification hazard that dictates *which* estimator to trust. The matched set is a crossed 41-solute $\times$ 17-solvent design, so any two pairs sharing a solute or a solvent share 51 of 102  $z^*$  coordinates exactly; the raw nearest neighbour of a pair is such a near-duplicate for 98% of pairs, which deflates the  $z^*$ -neighbour estimators anti-conservatively—the  $k$ -NN bound reads 0.40, but forbidding shared-

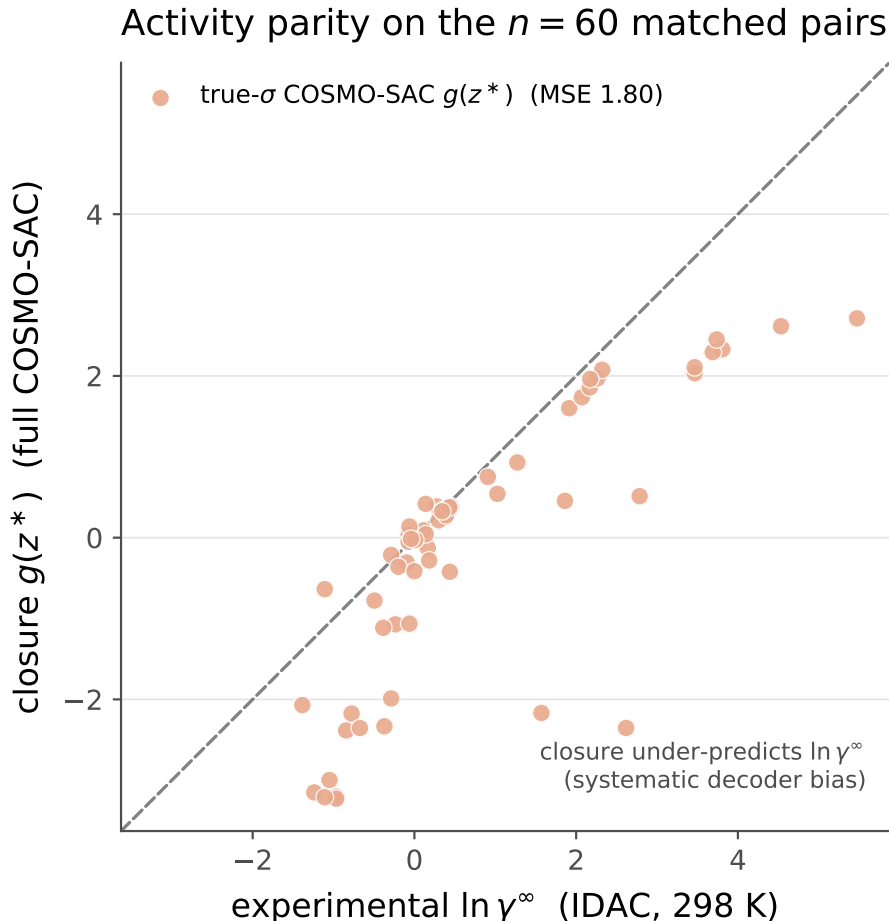


Figure 7: Activity-coefficient parity on the  $n=60$  matched pairs. The true- $\sigma$  COSMO-SAC closure  $g(z^*)$  (salmon) falls systematically below the diagonal at larger  $\ln \gamma_2^\infty$ , under-predicting the activity coefficient (MSE 1.80, full convention). This systematic downward bias is the visible form of the decoder misspecification  $B_{\text{closure}}$ ; for reference a flexible regressor of  $\ln \gamma_2^\infty$  on the same true profiles reaches the irreducible-variance floor (MSE  $\approx 0.40$ , our  $B_{\text{insuff}}$  estimator; §6.2), so the closure’s excess over that floor is  $B_{\text{closure}}$ .

solute/solvent neighbours *inflates* it to 1.83 as the neighbour gap takes over. Local conditional-variance estimators therefore *bracket*  $B_{\text{insuff}}$  across MSE/2 and cannot pin it, so we do not lean on the  $k$ -NN or random-forest out-of-fold values. We lean instead on the law-of-total-variance binning bound, which conditions  $m$  on the coarsened scalar  $g(z^*)$  rather than on  $z^*$ -neighbours: it is a valid upper bound on  $B_{\text{insuff}}$  (a binning of  $g(z^*)$  is a function of  $z^*$ ) and is *immune* to the shared-coordinate leakage. That bound is stable—it survives leave-one-solvent-out (17/17 folds), leave-one-solute-out (40/41), leave-one-pair-out (60/60), and adversarial deletion of the eight highest-error pairs. For confidence, a single-axis solvent-clustered bootstrap gives  $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.89$ ; a two-way (solute $\times$ solvent) cluster bootstrap that also charges for repeated solutes lowers this to  $P \approx 0.78$  with a 90% margin spanning zero—so we present the separation as a *supporting* measurement, likely but not certified at conventional confidence, not a stand-alone certificate. Stratifying the matched set by  $\ln \gamma_2^\infty$  under the same leakage-immune binning bound, the verdict holds in every stratum and the closure lower bound still increases with  $\gamma$  ( $0.95 \rightarrow 2.69$ ); but

Table 3: Closure–insufficiency decomposition on the true-input activity path. Bounds, not a point split (Lemma 3). Convention “full” = COSMO-SAC with the combinatorial term (size factor  $r_i = V_i/r_0$  on the true COSMO cavity volume, VT-2005); “res” = residual-only (the deployed default).  $B_{\text{insuff}}$  upper bounds are convention-independent estimates of  $\mathbb{E}[\text{Var}(m \mid z^*)]$ ; “kNN” is a direct high-dimensional estimate (biased upward), the others are law-of-total-variance / out-of-fold upper bounds. The *load-bearing* separation is the convention-independent  $B_{\text{closure}} \geq \text{MSE} - B_{\text{insuff}}^{\text{up}}$  row, which exceeds  $B_{\text{insuff}}$  under *both* conventions (1.27,  $0.85 > 0.62$ ); the assumption-free constant-offset (Jensen) bound separates only under “full” (0.72) and inverts under the deployed “res” ( $0.43 < 0.62$ ).

| Quantity                                                                                                                         | full                                     | res  |
|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------|
| $\text{MSE}_{\text{total}} = B_{\text{closure}} + B_{\text{insuff}}$ (exact)                                                     | 1.80                                     | 1.47 |
| $B_{\text{closure}}$ lower via $\text{MSE} - B_{\text{insuff}}^{\text{up}}$ (LOTV; <i>leakage-immune</i> , <i>load-bearing</i> ) | 1.27                                     | 0.85 |
| $B_{\text{closure}}$ lower bound (Jensen, offset <sup>2</sup> ; assumption-free; <i>full-only</i> )                              | 0.72                                     | 0.43 |
| $B_{\text{insuff}}$ upper (bin $g$ ; law of total variance)                                                                      | 0.53                                     | 0.62 |
| $B_{\text{insuff}}$ upper (RF / Ridge out-of-fold)                                                                               | 0.56/0.62                                | same |
| $B_{\text{insuff}}$ (kNN-in- $z^*$ , convention-indep., biased up)                                                               | 0.40                                     | 0.43 |
| label noise $\sigma_\eta^2$ (IDAC, unestimable here)                                                                             | assumed 0 (no replicates in matched set) |      |

the higher- $\gamma$  margin is thin and estimator-dependent (the random-forest bound flips there), so we read the shared cause with the all- $\gamma$  paradox as a conjecture supported by the *direction* of the in-regime signs, not a measured magnitude (App. C). Under the full COSMO-SAC convention the separation *sharpens* to an assumption-free constant-offset bound: a constant decoder offset alone accounts for  $B_{\text{closure}} \geq 0.72$  (Fig. 8), exceeding every  $B_{\text{insuff}}$  upper bound ( $\leq 0.62$ ), and subtracting the insufficiency bound gives  $B_{\text{closure}} \geq 1.24 > 0.62$  ( $\sim 2\times$ ). That constant-offset bound is convention-specific and *inverts* under the deployed residual-only default ( $0.43 < 0.62$ ; hedge 1), which is exactly why we headline the convention-independent bound above rather than the constant offset. Label noise is unestimable in this replicate-free set (the matched IDAC pairs carry no repeat measurements at 298 K, so an inter-source floor cannot be formed as it was for solubility) and assumed zero; the bounds we rely on are invariant to zero-mean noise. A *systematic* between-method offset in the compiled IDAC labels (gas–liquid chromatography, ebulliometry, and dilutor techniques carry known method-dependent bias, not zero-mean scatter) is *not* covered by that invariance and would inflate  $B_{\text{closure}}$ ; we cannot estimate it on this replicate-free set and flag it as an untested exposure of the attribution. Because a replicate floor is unavailable, we test exposure to the highest-scatter data directly: the separation *sign* survives deleting any single pair (60/60) and trimming the eight highest-error pairs—which include the physically suspect associating outliers (e.g. 1-hexanamine in methanol,  $\ln \gamma^\infty = +2.62$  against its reverse  $-0.97$ )—though under that trim the estimated  $B_{\text{closure}}$  lower bound falls  $\sim 58\%$  ( $0.91 \rightarrow 0.38$ ). The closure-dominance *sign*, not its magnitude, is leverage-robust.

**The closure defect is low-rank and correctable.** That  $B_{\text{closure}}$  dominates locates the ceiling in the map; it does not say the map is *fixably* wrong. It is. Fitting a small structure-preserving residual to the fixed COSMO-SAC exchange-energy kernel— $\Delta = B \text{diag}(a) B^\top$  over the  $\sigma$ -grid, symmetric and zero-initialised, the same object a learned closure-correction (Arm C) would carry—on the *true* VT-2005 profiles removes 56% of the closure error out-of-sample (five-fold cross-validated, activity  $\text{MSE } 1.47 \rightarrow 0.64$  held-out), and does so at *rank one* (52 parameters): rank 2 adds nothing, rank 3 begins to over-fit. (At 52 parameters against  $n=60$  this held-out reduction is provisional

Closure-insufficiency bounds, deployed residual-only COSMO-SAC ( $n = 60$ ):  $B_{\text{closure}}$  dominates (leakage-immune bound)

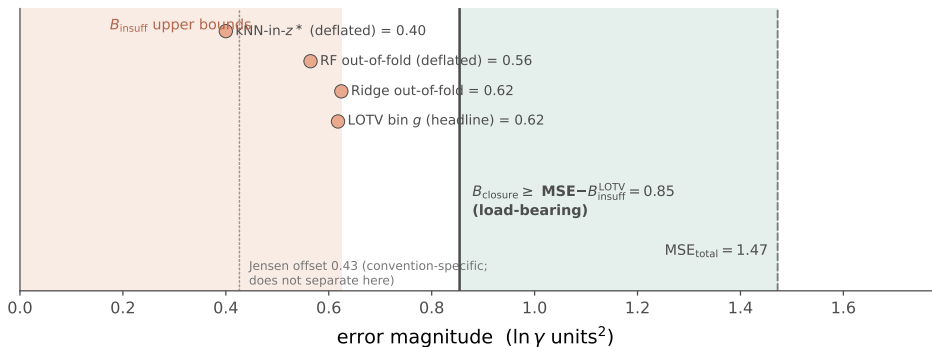


Figure 8: One-sided bounds on the closure-insufficiency split under the *deployed* residual-only COSMO-SAC convention—the configuration the model actually runs. The load-bearing, convention-independent bound  $B_{\text{closure}} \geq \text{MSE} - B_{\text{insuff}}^{\text{LOTV}} \approx 0.85$  exceeds every valid  $B_{\text{insuff}}$  upper bound (salmon region); the white gap is a separation that needs no smoothness assumption, and the teal region is the certified closure share of the total error ( $\text{MSE} = 1.47$ ). We headline the law-of-total-variance (LOTV) binning bound because it conditions on the scalar  $g(z^*)$  and is immune to the near-duplicate leakage that deflates the  $z^*$ -neighbour estimators ( $k$ -NN, RF; §6.2). The assumption-free Jensen constant-offset bound is convention-specific and *inverts* here ( $0.43 < 0.62$ , dotted), which is why we do not lean on it.

pending a larger matched set, and we have not yet separated the kernel correction from a per-solvent mean-offset removal.) So  $B_{\text{closure}}$  is not diffuse model error but a low-dimensional, correctable defect of the map—addressable with a handful of parameters placed on the closure, whereas the same error is only made worse by feeding the inputs truthfully (§6.1) and untouched by an output residual (§6.5). This is the constructive complement to the paradox, on the true-input set where the closure error lives; whether an end-to-end trained model realises the same gain is a compute-gated question we leave open. (The map correction and the input compensation of §6.3 occupy different  $\sigma$ -grid footprints—they are two routes to the same defect, not one.)

**Convention-robust corroboration (Tier 2).** The genuinely convention-robust fact is that adding the physically *more* complete Staverman–Guggenheim combinatorial term *increases* error (activity  $\text{MSE } 1.47 \rightarrow 1.80$ )—what a misspecified map does, not what missing information would. This is robust to the size-factor’s volume basis: recomputing the combinatorial term on the true COSMO cavity volumes (VT-2005  $V_{\text{cosmo}}$ ) rather than an RDKit molar volume leaves it essentially unchanged (Jensen  $0.714 \rightarrow 0.717$ ,  $\text{MSE } 1.793 \rightarrow 1.800$ ), because the term depends on molecular-size *ratios* in which a near-uniform volume rescaling cancels. The increase is, however, concentrated on size-asymmetric pairs (its magnitude correlates with  $|\ln(V_2/V_1)|$  at  $r=0.33$  and is  $11\times$  larger in the high-asymmetry half), i.e. it is the documented miscalibration of the *unmodified* combinatorial term for size-asymmetric mixtures—the reason modified-Staverman–Guggenheim and free-volume corrections exist—so we read it as corroboration of closure misspecification, not an independent second effect. The residual (deployed) error is itself localized, and it is not an implementation artifact: our differentiable layer reproduces the NIST benchmark COSMO-SAC-2002 [2] to RMSE  $0.003 \ln \gamma$  on identical VT-2005 profiles (0.005 even in the associating regime, max 0.02), and self-solvation  $\ln \gamma^\infty(i, i)$  is exactly zero. Yet 90% of the squared error falls on strong hydrogen-bond-donor solvents while acceptor-only solvents are near-exact (mean squared error 0.09)—placing

$B_{\text{closure}}$  in COSMO-SAC’s documented single-parameter hydrogen-bonding kernel, a real but *known* limitation rather than an idiosyncrasy of our differentiable layer. A second observation is consistent but is *not* independent of the decomposition: a flexible regressor of  $m$  on  $z^*$  fits the activity target  $\sim 4\times$  better than the fixed true-input closure—but that regressor is exactly our convention-independent  $B_{\text{insuff}}$  estimator ( $\approx 0.4$ ), so “ $4\times$  better” restates that the closure sits far above the irreducible-variance floor (that  $B_{\text{closure}}$  is large) rather than corroborating it by an independent route. We therefore lean on the dimension-robust random-forest / ridge out-of-fold and binning bounds and on the combinatorial effect, which agree that  $B_{\text{closure}}$  exceeds  $B_{\text{insuff}}$ .

**The ceiling moves with closure fidelity: a standard-closure control.** The 2002 closure-dominance above is not a general fact about physical inputs; it is specific to the deployed map’s single-parameter hydrogen-bonding kernel, and the field already has better-parameterized closures built for exactly that defect. We run the NIST reference implementation of COSMO-SAC-2010 (donor/acceptor-typed HB) and -dsp on the same matched pairs, now under the reference Delaware (UD) 3-profile database—on which the reference 2002 map scores  $\text{MSE} = 1.26$ , against 1.47 for our deployed residual-only layer on the VT-2005 profiles used elsewhere. The better-parameterized closures cut the error to  $\text{MSE} = 0.77$  (2010) /  $0.79$  (dsp,  $n=57$ ): a  $\sim 70\%$  reduction of  $B_{\text{closure}}$  ( $0.70 \rightarrow 0.20$  at point) against the unchanged  $B_{\text{insuff}}$  floor. Under two-way solute $\times$ solvent clustering this *dissolves the closure-dominance certificate*— $P \approx 0.79$  that  $B_{\text{closure}}$  no longer exceeds  $B_{\text{insuff}}$ . We do not claim a certified flip to the inputs: we hold only an *upper* bound on  $B_{\text{insuff}}$ , so “the inputs bind” would require a lower bound. That reading does hold for any true  $B_{\text{insuff}} > \text{MSE}_{2010}/2 \approx 0.38$ , which is below every  $B_{\text{insuff}}$  estimate we have (0.56–0.63; only the downward-biased  $k$ -NN floor 0.40 approaches it). The decision-relevant reading is thus not “the closure always dominates” but “on a low-fidelity closure it does, and the fix is a better closure, not truer inputs”—closure fidelity is the second axis of the map (§7).

**Scope and hedges.** An adversarial audit sharpened the scope without inverting the verdict. The constant-offset bound is convention-specific—it inverts under the deployed residual-only default, where the convention-independent bound ( $B_{\text{closure}} \geq 0.85 > 0.62$ ) and the combinatorial fact carry the claim instead; the closure error is chiefly a between-solvent bias (a solvent-conditional lower bound  $B_{\text{closure}} \geq 0.89$ ); the solvent-clustered confidence on the constant-offset separation is  $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.85$  (0.83 vs the random-forest bound, 0.89 vs the law-of-total-variance bound); and the measurement is confined to the low- $\gamma$ , 298 K, VT-2005-matchable corner, so the numeric ceiling is not claimed to transfer to the finite-composition regime. The four hedges are stated in full in App. C.

### 6.3 Why grounding hurts on this anchor: the end-to-end latent is a compensating surrogate

This is the *mechanism* behind the low-fidelity, end-to-end corner of the map (§7): when a  $\sigma$ -profile head is trained *end-to-end* through a fixed, misspecified closure—never supervised toward the true profile—it is under pressure not to be physically faithful but to *cancel* the closure’s error. The external ground truth that lets us measure the ceiling (§6.2) also lets us test this directly, comparing the predicted profile  $\hat{\sigma}$  against the true VT-2005 profile  $\sigma$  molecule by molecule ( $n=44$ ).

**The learned profile is a systematic, low-rank, transferable surrogate—confound-free.** We use the cleanest test available: a *single* grounded model checkpointed before and after solubility fine-tuning (the  $\sigma$  head unfrozen), so architecture and initialisation are held fixed and only the

objective differs. The grounded base recovers the true profile to within 33% ( $\|\hat{\sigma}_{\text{grounded}} - \sigma\|/\|\sigma\|$ ), so the intermediate *can* be physical; solubility fine-tuning then drives it 45% off the grounded profile, for a total departure from truth of 53%. The drift is not per-molecule noise: its first two principal components carry 73% of the variance (isotropic noise over 51 bins would give  $\approx 4\%$ ), and a single affine distortion  $D$  fit on half the molecules predicts the drift on the held-out half  $3.3\times$  better than identity out-of-sample. So  $\hat{\sigma}$  is a systematic, low-rank, *transferable* closure-compensating surrogate—one generalizable way the intermediate “lies” to the closure, shared across chemistry, driven by the fitting objective rather than by architecture or a lack of grounding. (The magnitude develops over convergence—a short run reached only  $\sim 16\%$ ; the converged control above supersedes it. Earlier confounded and two-model analyses over-stated the departure at  $\sim 82\%$  and the transfer at  $16\text{--}26\times$ ; we report the confound-free numbers.)

**Dominant mode and consequence.** The dominant drift mode (Fig. 10) transfers area *out of* the nonpolar bins ( $\sigma \approx 0$ ) and *into* the polar wings: the network makes molecules look more polar than they are, inflating the segment interactions the misspecified exchange kernel underweights and thereby cancelling part of  $B_{\text{closure}}$ —the input compensating the map, exactly as the non-identifiability of the split predicts (Lemma 1). This re-explains why grounding hurts *here* (§6.1): the pipeline is calibrated to the compensated  $\hat{\sigma}$ , not to physics, so substituting the true  $\sigma$  breaks the compensation. It also means the model’s “physical” intermediate is not the molecule’s charge distribution but a surrogate, consistent with known cautions that the interpretability of physics-grounded latents is not guaranteed under misspecification [3, 27]; what is specific here is the *structure*—the surrogate is low-rank and transferable, not diffuse.

**The structure is not the closure’s linear response.** A natural hope is that this structure is simply the closure’s first-order inverse: the smallest input change that cancels the closure’s discrepancy,  $\hat{\sigma} - \sigma \approx J^+(m - g(\sigma))$ , with  $J = \partial g/\partial \sigma$  the closure Jacobian and  $m$  the observed  $\ln \gamma_2^\infty$ —a drift that would then be *predictable a priori* from  $J$ , before any solubility training. It is not. Testing this on the matched set (per molecule, the compensation solved in the span of its own pairs’ Jacobian rows), the measured drift shows no alignment with  $J^+(m - g)$ : the per-molecule cosine is centred on zero (median  $-0.00$ ), a smoothness-matched control—aligning each molecule’s drift against *other* molecules’ predicted drift, all equally smooth—shows no molecule-specific gap (own vs. other cosine  $-0.00$  vs.  $-0.01$ , permutation  $p=0.17$ ), and the drift’s energy inside each molecule’s own Jacobian subspace (5.5%) is no larger than inside other molecules’ (5.0%; sign-test  $p=0.76$ ); the null is unchanged when pairs are weighted by their residual or restricted to the largest residuals, where the compensation pressure is strongest.<sup>2</sup> So the surrogate is systematic and transferable yet *not closure-geometric*: it is an affine distortion in  $\sigma$ -space (Fig. 9), not the closure’s first-order inverse. This sharpens the interpretability caution rather than softening it—the intermediate’s departure from physics cannot be undone by inverting the closure, so the true  $\sigma$  is not recoverable from the misspecified pipeline even in principle.

**The contrast that fixes the sign.** This is why the sign of grounding is not universal. When the  $\sigma$  head is instead *supervised* toward the true profile, the intermediate stays close to physical and true profiles *help*: the recent TeNNet-SAC model [35] supervises its  $\sigma$ -profiles toward quantum-chemistry values and, once fine-tuned end-to-end, outperforms COSMO-SAC. Our end-to-end head,

<sup>2</sup>The residual COSMO-SAC closure runs a segment-activity fixed-point iteration, so its *autograd* Jacobian is numerically unstable ( $\|\partial g/\partial \sigma_{\text{solvent}}\| \sim 10^{15}$  at infinite dilution);  $J$  is therefore computed by finite differences on the converged closure.

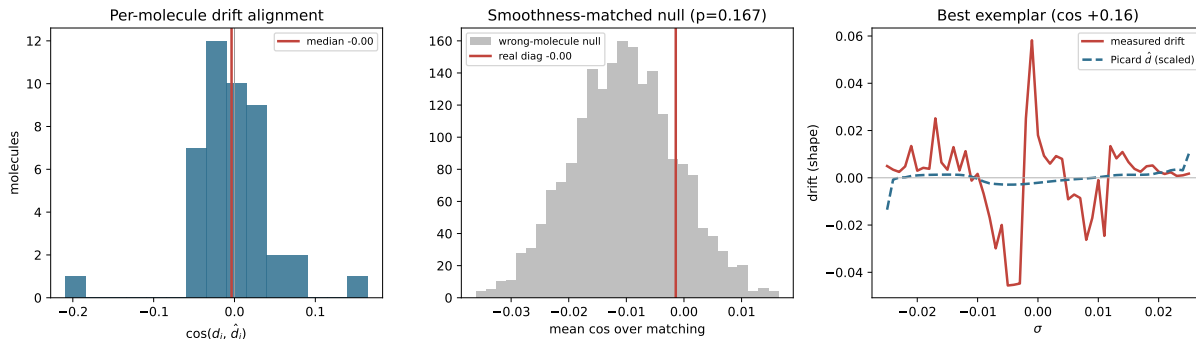


Figure 9: The compensating drift is *not* the closure’s linear-response inverse. *Left*: per-molecule cosine between the measured drift  $\hat{\sigma} - \sigma$  and the first-order closure compensation  $J^+(m - g)$  is centred on zero (median  $-0.00$ ). *Middle*: the smoothness-matched wrong-molecule control—each molecule’s drift aligned against every other molecule’s predicted drift, all equally smooth—places the true diagonal squarely inside the null ( $p=0.17$ ), so any weak alignment is generic smoothness, not molecule-specific closure geometry. *Right*: the best-aligned exemplar, showing that even the strongest case is not tracked by the Picard prediction. The surrogate is structured but not readable from the closure.

un-supervised and pushed through a low-fidelity closure, becomes a surrogate and true profiles *hurt*. Latent supervision is one of the two axes that govern the sign (§7). The scope limit is the low- $\gamma$ ,  $n=44$  VT-2005 matched set.

## 6.4 Closure fidelity sets the true-input penalty

To isolate the mechanism we built a controlled synthetic experiment in which only the closure’s fidelity varies (input quality, sample size, and estimator held fixed). A teacher draws a latent  $z^*$  and a target through a known map  $T$ ; a fixed candidate closure  $g_F$  reconstructs it under a tunable misspecification of scalar fidelity  $F$  ( $F = 1$  recovers  $T$  exactly, smaller  $F$  pushes  $g_F$  away). The true-input penalty  $\Delta R^2 = R_{\text{oracle}}^2 - R_{\text{head}}^2$  is  $B_{\text{closure}}$  made visible—the free head recovers  $\mathbb{E}[m \mid z^*]$  while  $g_F$  is the misspecified map. As  $F$  falls the penalty grows monotonically and without exception: averaged over six misspecification forms (linear, quadratic, cubic, absolute-value, sinusoidal, cross-term)  $F=1.00$  gives  $\Delta R^2 \approx 0$ ,  $F=0.38$  gives  $\approx -0.33$ , and an over-rotation stress test ( $F=-0.50$ ) gives  $\approx -2.0$  (Fig. 11); the ordering is identical across four unrelated teacher families (linear, monotone-nonlinear, kinetics-exponential, PDE-field), so the effect is a property of fidelity magnitude, not of any one domain. Placing the real system on the dial only illustratively (the real-closure-to- $F$  map is a synthetic construction, not a calibrated measurement), the COSMO-SAC closure over predicted  $\sigma$ -profiles falls at  $F \lesssim 0.4$ : its solubility  $\Delta R^2 \approx -0.36$  is what a closure of this fidelity predicts, consistent with  $B_{\text{closure}}$  dominating  $B_{\text{insuff}}$ .

The same engine shows why the grounding gap is a symptom, not a certificate (§4.3). Holding the closure *well specified* ( $B_{\text{closure}} = 0$ ) and varying how much target-relevant information  $z^*$  discards: when  $z^*$  is sufficient,  $\Gamma$  tracks  $B_{\text{closure}}$  as fidelity varies (Theorem 1’s proxy is exact); when  $z^*$  is lossy but the closure is correct,  $B_{\text{closure}} = 0$  throughout yet  $\Gamma > 0$  grows with the discarded variance (Fig. 12). A positive grounding gap can thus be pure input insufficiency, not closure error—so closure-dominance is certified by the convention-independent decomposition, and  $\Gamma$  only corroborates it.

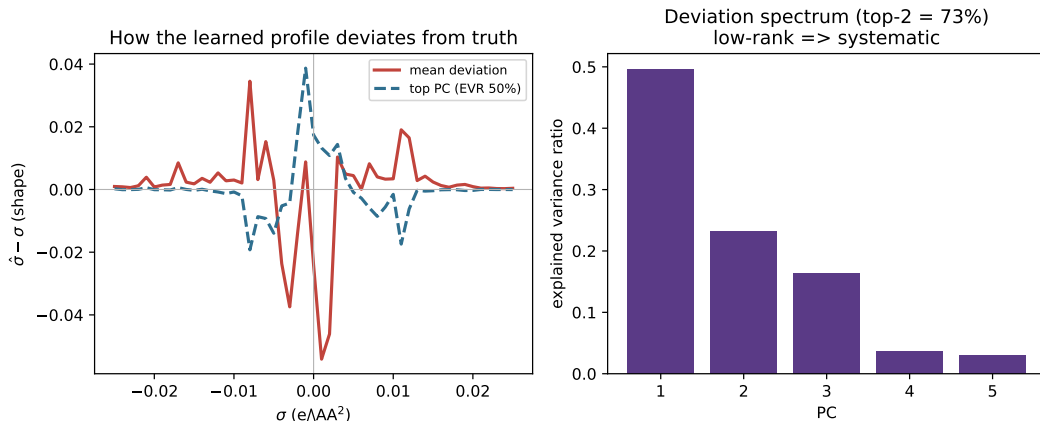


Figure 10: The learned  $\sigma$ -profile is a systematic closure-compensating surrogate, not the physical profile. *Left*: the mean deviation  $\hat{\sigma} - \sigma$  (and its dominant PCA mode) transfers area out of the nonpolar bins ( $\sigma \approx 0$ ) into the polar wings—the network makes molecules look more polar than they are, inflating the segment interactions the misspecified closure under-weights. *Right*: the deviation spectrum is low-rank (top-two modes 73%; noise would be flat at  $\approx 4\%$ ), i.e. the distortion is systematic. A single affine distortion transfers  $3.3\times$  better than identity out-of-sample, so  $\hat{\sigma}$  is a transferable closure-compensating surrogate of  $\sigma$ , not  $\sigma$  with noise (confound-free one-model control,  $n=44$  matched VT-2005 molecules).

## 6.5 The ceiling is not fixable downstream (Gate B)

If the closure binding were merely a bad point estimate, a learned output-space correction on top of the physics prediction should recover accuracy. It does not. An additive bounded residual with a learned confidence gate ( $\hat{y} = \hat{y}_{\text{phys}} + (1-c)r$ , with  $y = \ln x_2$ ) leaves the valve effectively shut (applied residual  $\approx 0$ ). Forcing the gate open removes the systematic *bias* ( $+0.45 \rightarrow +0.06$ ) but recovers little *accuracy* ( $R^2$   $0.236 \rightarrow 0.274$  on the same rows, a marginal rise that leaves the fit far short of usable). This is the classical calibration-versus-accuracy separation: when the upstream map is misspecified, a monotone or affine post-hoc map can fix reliability but not resolution. Recalibration buys calibration without buying accuracy, which is what we expect if the closure, rather than the estimate, is the binding constraint.

## 6.6 A second chemical closure: pKa through a Hammett LFER

The machinery is not specific to COSMO-SAC. In a second real domain—aqueous  $\text{p}K_a$  of substituted aromatics—the fixed closure is a Hammett linear-free-energy relationship  $g(\sigma) = \text{p}K_{a,0} - \rho \sigma$  over the Hammett substituent constant  $\sigma$  (*not* the  $\sigma$ -profile of §3), for which Hansch–Leo tabulated constants supply the external oracle. The two classic LFER failure modes fall cleanly onto the two channels of Lemma 3: *resonance saturation* (a curvature the linear closure cannot represent) lands in  $B_{\text{closure}}$ , while *ortho/steric* effects, absent from the tabulated constants, are information the intermediate lacks and land in  $B_{\text{insuff}}$ . On the OPERA experimental compilation [18] this yields a *measured* fidelity sweep: the clean meta/para series ( $n=161$ ) is well specified (RMSE 0.63,  $B_{\text{closure}}$  below the insufficiency floor), while the ortho/heteroaromatic regime ( $n=1008$ ) degrades to RMSE 2.4, input-insufficiency dominated—the near-exact-to-degraded transition, measured rather than asserted. A trained physics arm does beat its matched DirectGNN control overall (1.63 vs 1.84



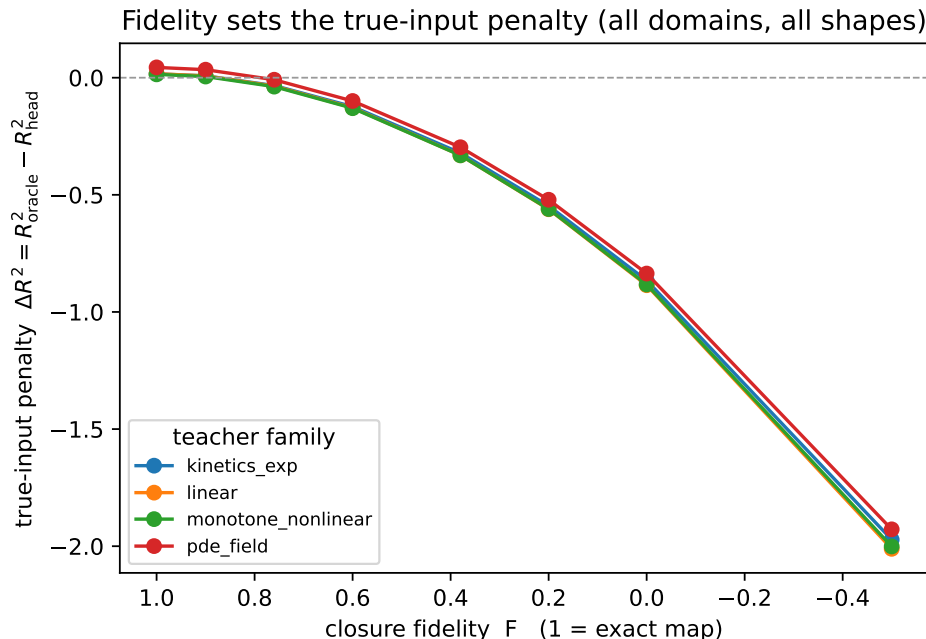


Figure 11: Reproducible closure-fidelity dial (`run_fidelity_dial.py`). As the closure is made more misspecified (fidelity  $F$  decreasing from 1), the true-input penalty  $\Delta R^2 = R^2_{\text{oracle}} - R^2_{\text{head}}$  grows monotonically, and the same ordering holds across four non-chemistry teacher families (linear, monotone, kinetics-exponential, PDE-field) and six misspecification functional forms. The effect is a property of the fidelity magnitude, not of any one domain or deformation. The real COSMO-SAC closure sits well inside the misspecified regime ( $F \lesssim 0.4$ )—the sign and rough magnitude of the empirical paradox, not a precise fidelity value.

pKa MAE), but a stratified check shows the gain concentrates in the ortho regime, where the *freely* learned  $\sigma$  absorbs information the tabulated constant lacks and a per-scaffold prior is strong—a permissive-closure, not a clean closure-fidelity, effect—so we read it cautiously and do not lean on it. The full construction, the semi-synthetic estimator validation, the measured OPERA sweep, and the trained comparison with its stratification are in App. G (Fig. 16).

## 6.7 Grounding does not pay off in the low-data regime either

The strongest a-priori case for a physics prior is data scarcity: with few training molecules, an inductive bias toward the correct physical factorization should beat a data-hungry black box even when the two tie at full data. We test this directly. Subsampling the training set *by solute* at fractions 0.05–1.0 (the regime that governs scaffold transfer), we train the physics-grounded model and DirectGNN on the identical subsample and evaluate both on the fixed scaffold test (Table 4; seed 42, full-corpus run). The physics arm is *worse at every fraction*, and—against the hypothesis—the gap is *smallest* at the least data ( $\Delta\text{MAE} = +0.08$  at 5%) and *grows* with more (+1.0 at full); its  $R^2$  is negative throughout. We do not map this to Prop. 1: the physics arm is markedly more under-converged than the control at this light per-fraction budget, so the within-budget gap reflects optimization difficulty as much as inductive bias. We do not over-read this: the sweep is single-seed (seed 42), the low-data gap is small (+0.08 at 5%), and the physics-arm MAEs are non-monotone across fractions, so we take it as *consistent with*—not a sharp confirmation of—the sign rule, and

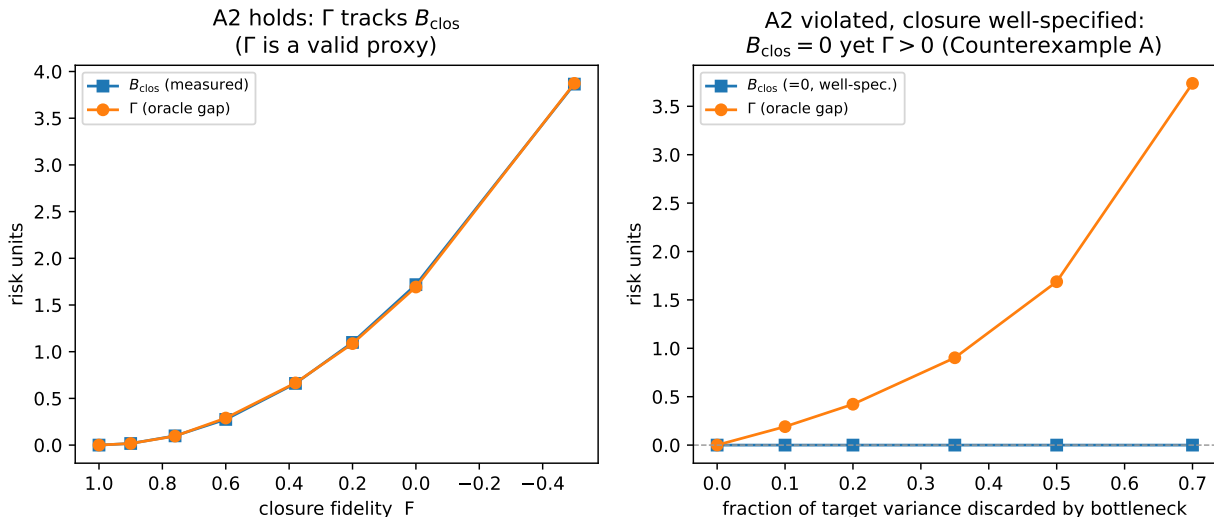


Figure 12: The grounding gap  $\Gamma$  is a symptom, not a certificate (§4.3; `run_fidelity_dial.py`). *Left*: when the intermediate is sufficient (A2 holds),  $\Gamma$  tracks the closure bias  $B_{\text{closure}}$  as fidelity varies, so Theorem 1’s proxy is valid. *Right*: with a *well-specified* closure ( $B_{\text{closure}} = 0$ ) but a lossy intermediate (A2 fails),  $\Gamma > 0$  and grows with the discarded variance—a positive grounding gap from pure input insufficiency (the worked counterexample in the proof of Theorem 1, App. B). Hence certification of closure-dominance rests on the decomposition, which needs neither assumption.

leave multi-seed error bars at each fraction to future work. Even in the regime built to favor it, grounding does not buy accuracy; if anything, more data lets the black box pull further ahead. The one metric that inverts is top-1 solvent ranking, higher for the physics arm at 5% (0.42 vs 0.23)—consistent with the ranking-versus-accuracy split of §6.8—but that decision-quality signal does not translate into lower error. This is the last regime in which a physics prior should plausibly have helped; that it does not closes the accuracy case and leaves the structural account of §4 and §6.2—a non-identifiable split fed through a misspecified closure—as the operative explanation.

## 6.8 Supporting diagnostics

The interpretive claims of the preceding sections rest on a set of controlled diagnostics, each run under a named protocol. We report them here with their exact numbers, including the results that run against the physics-informed hypothesis.

**Aleatoric noise floor.** Repeated solubility measurements of the same solute in the same solvent disagree across laboratories, and that disagreement sets a floor below which no model can be expected to fit. Measuring inter-source spread on [13], the mean absolute disagreement is approximately  $0.15 \ln x_2$  units for pairs backed by two sources ( $n = 3948$  groups), rising to 0.31 for groups backed by three or more sources ( $n = 349$ ); the corresponding medians are smaller still (0.03 and 0.09). The scaffold-split MAE we report ( $\approx 1.7$  to 1.9) sits well above this band, so label noise does not account for the accuracy ceiling. The floor is small; the ceiling is not a data-quality artifact.

**The encoder is transfer-limited, not expressivity-limited.** To separate what the frozen graph embedding can represent from what it transfers out of distribution, we fit a linear probe

Table 4: Data-efficiency sweep (train subsampled by solute, fixed scaffold test, seed 42). The physics-grounded arm trails DirectGNN at every fraction; the gap is smallest at the least data and large thereafter.  $\Delta\text{MAE} = \text{MAE}(\text{physics}) - \text{MAE}(\text{direct}) > 0$  means physics is worse. Absolute errors here exceed the headline runs (a lighter per-fraction budget); the valid signal is the within-budget physics–direct gap.

| train fraction<br>(by solute) | MAE     |        | $\Delta\text{MAE}$<br>(phys–dir) | $R^2$   |        |
|-------------------------------|---------|--------|----------------------------------|---------|--------|
|                               | physics | direct |                                  | physics | direct |
| 0.05                          | 2.28    | 2.20   | +0.08                            | −0.09   | +0.07  |
| 0.10                          | 2.44    | 2.07   | +0.37                            | −0.25   | +0.15  |
| 0.25                          | 2.93    | 1.81   | +1.12                            | −0.75   | +0.38  |
| 0.50                          | 3.12    | 1.98   | +1.14                            | −0.98   | +0.22  |
| 1.00                          | 2.98    | 1.97   | +1.01                            | −0.79   | +0.22  |

from the encoder representation onto RDKit descriptors and compare in-distribution and held-out fit. The median train  $R^2$  is 0.758 against a test  $R^2$  of 0.448 ( $n = 700$  solutes in each partition), a gap of 0.31. Individual descriptors split sharply: FractionCSP3 transfers well (test  $R^2 = 0.96$ ) while HeavyAtomCount does not (0.23). The embedding is expressive enough in sample; it is the generalization of that expressivity across scaffolds that fails. Swapping in a new graph encoder is therefore unlikely to lift out-of-distribution accuracy, and the useful levers are pretraining and coverage rather than architecture.

**Temperature extrapolation.** A structured van’t Hoff activity class (log-solubility affine in  $1/T$ ) encodes the correct temperature dependence of  $\ln \gamma_2$  by construction, and this shows up when the split withholds temperature rather than structure. Training on measurements at  $T \leq 310$  K and testing at  $T \geq 330$  K ( $n_{\text{test}} = 3343$ ), the structured class attains a high-temperature MAE of 0.37 against 1.29 for a random forest over Morgan fingerprints plus temperature (in  $\ln x_2$  units), consistent with the temperature-aware  $\ln \gamma_2^\infty$  modelling of [26]. This is a property of the structured hypothesis class, not a performance claim we make for the trained model; the point is that the functional form carries temperature extrapolation that a descriptor-plus-temperature regressor does not.

**Ranking and calibration.** Absolute error and decision quality come apart. Despite mediocre MAE, the physics path orders candidate solvents usefully (Table 5; nDCG@3  $\approx 0.73$  across  $n = 826$  solute groups). Uncertainty, by contrast, is badly miscalibrated in the native model—a nominal-90% interval covers only PICP@90 = 0.13 of held-out points—but split-conformal wrapping repairs it to 0.92 empirical coverage at the same nominal level. The ranking signal and the post-hoc calibrated intervals are the practically usable outputs, even where the point predictions are not competitive.

## 7 Discussion

The result is a cautionary measurement for physics-informed modeling: grounding a model on truer physical inputs is only as good as the closure that consumes them, and when that closure is a fixed, mildly misspecified map, better inputs surface its bias rather than help. The external latent does not make the physics path win. What it buys is *attribution*: on the infinite-dilution, low- $\gamma$  subset

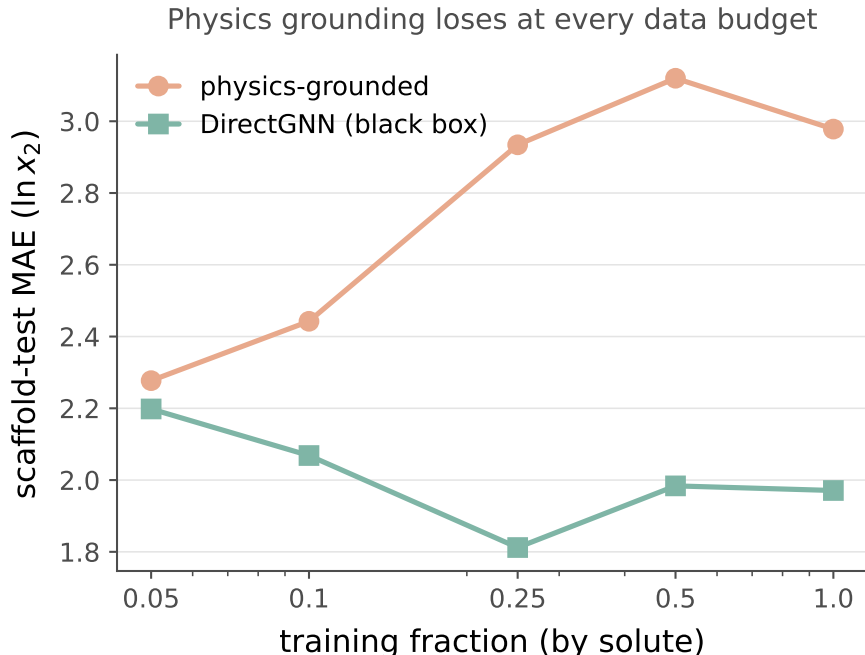


Figure 13: Data-efficiency curve. The physics-grounded arm (salmon) trails the DirectGNN black box (teal) at every training-data fraction; the gap is smallest at the least data and large thereafter—the opposite of the low-data advantage a physics prior is supposed to provide.

Table 5: Solvent-ranking quality of the physics path on the scaffold test ( $n = 826$  solute groups).

| Metric          | Value |
|-----------------|-------|
| Spearman $\rho$ | 0.511 |
| Top-1 accuracy  | 0.477 |
| Kendall $\tau$  | 0.436 |
| nDCG@3          | 0.732 |

we can measure, the ceiling sits in the decoder ( $B_{\text{closure}}$ ), with the inputs ( $B_{\text{insuff}}$ ) the less likely source and label noise unestimable but assumed zero (and unable to manufacture the constant offset the bounds rest on). We do not claim this numeric attribution transfers unchanged to the finite-composition, high- $\gamma$  regime the solubility paradox itself spans; that the solubility paradox shares the same closure-misspecification cause is a conjecture supported by in-regime signs—the more-physics-worse effect on the full  $n=5608$  set and the chemistry map—not a measured fact. For practice this relocates the lever. Within this closure family, effort spent on richer inputs or a better encoder is mis-targeted; the payoff is in the closure itself, for example a corrected or learned residual on the map rather than on its output (which §6.5 shows is inert).

**The accuracy case is broad, with an explained boundary.** That the physics prior does not improve accuracy is a pattern with no confirmed exception *on accuracy*: the grounded model trails the matched black box on the full corpus (and the external state-of-the-art (SOTA), FastSolv and SolProp retrained on our corpus, does no better—though on a by-solute split that bounds task difficulty rather than ranking head-to-head; App. D), a (dose-limited) attempt to ground the crystal

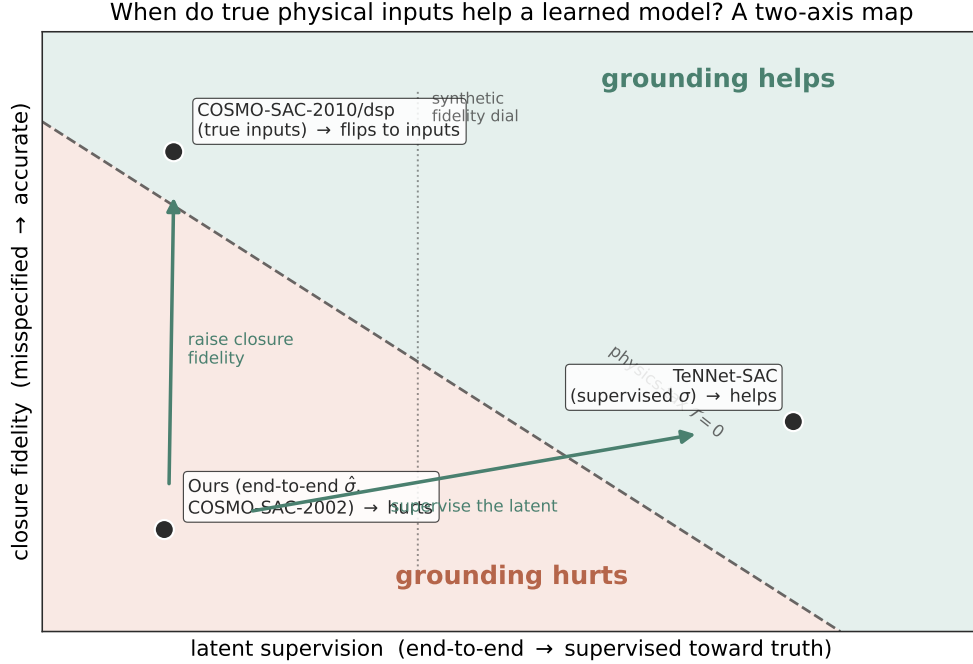


Figure 14: The grounding map. Whether feeding a learned model its true physical inputs helps or hurts is governed by two axes—how the latent is trained (supervised toward truth vs. shaped end-to-end through the fixed closure) and the closure’s fidelity. Grounding hurts only in the low-supervision, low-fidelity corner, where the end-to-end latent becomes a compensating surrogate (§6.3). Our COSMO-SAC-2002 model sits there; raising closure fidelity (COSMO-SAC-2010/dsp, §6.2) or supervising the latent (TeNNNet-SAC) crosses into the “helps” region. The boundary is the physics-tax sign  $\mathcal{T} = 0$  (Prop. 1); the synthetic dial (§6.4) sweeps the fidelity axis. Schematic; anchor placements are qualitative.

factor on external labels did not move the ceiling (§F.2, inconclusive), and—in the case built to favour a physics prior—it loses at *every* training-data fraction, including the low-data regime where an inductive bias should help most (§6.7). The one axis on which the physical *structure* does help is temperature extrapolation—but as a hypothesis-class property that the trained models do not exploit (§6.8), a partial counterexample we flag rather than hide. Read together these point to a *structural* ceiling: the crystal/activity split is non-identifiable from solubility alone (Lemma 1), and the activity closure is misspecified where we can measure it (§6.2)—a measurement that feeds true inputs through the fixed closure and so cannot be a training artifact—so within this family better inputs, more data, and auxiliary grounding do not recover an accuracy the factorization cannot express. The contribution is accordingly not an accuracy result but a *map* of when grounding a learned latent in true physics helps or hurts: the externally grounded decomposition *locates* the ceiling (measured, not proved), a standard higher-fidelity closure moves it and dissolves the closure-dominance certificate, and—where the closure binds and the latent is end-to-end—the “physical” intermediate behaves as a compensating surrogate; the two governing axes are closure fidelity and latent supervision (Fig. 14). What a misspecified closure costs is thus not only accuracy but the interpretability the bottleneck appeared to buy.

**When a physics prior must lose.** The tax rule (Prop. 1) turns these negatives into a phase picture rather than a single verdict (Fig. 15), and says *when* to expect them. A fixed-closure prior helps only when the estimation variance it saves exceeds its asymptotic bias  $\Delta_\infty$  (equal to  $B_{\text{closure}} - \Gamma$  only under A2 with an asymptotically-Bayes control, otherwise read off the large- $n$  asymptote); both are measurable. The variance benefit is largest at small  $n$  and vanishes as data accumulate or as the black box nears the aleatoric floor, whereas  $\Delta_\infty$  is set by the closure and does not. A prior is therefore predicted to hurt precisely in the mature-data, low-noise, well-covered regime—exactly where our data-efficiency null (a positive tax at every budget, §6.7) and the novelty inversion (the tax heaviest on the best-covered solutes, §F.3) place this system, and where much of modern machine learning operates. The synthetic dial (§6.4) and the pKa closure (§6.6) trace the two governing axes—closure fidelity and intermediate sufficiency—directly. We claim no universal boundary; we *propose*, but do not here cleanly validate, a diagnostic for the sign of the physics tax: measure  $B_{\text{closure}}$  through the decomposition, read  $\Delta_\infty$  off the large- $n$  asymptote, and the sign should follow—a prediction our own tests (single-seed, under-converged; §6.7) support only weakly. In this sense the failure is not idiosyncratic to solubility; it recurs whenever a misspecified closure is coupled to an expressive encoder on abundant, low-noise data.

**Limitations.** (i)  $B_{\text{insuff}}$  is bounded, never point-identified: there are no true replicate measurements at fixed  $z^*$ , so the insufficiency estimates lean on a smoothness assumption. The constant-offset (Jensen) bound is smoothness-free but assumes  $m$  and  $g$  share the  $\ln \gamma_2^\infty$  (mole-fraction, Lewis-Randall) reference state; we check this two ways—the closure is nearly exact within the largest single-solvent stratum (pyridine, MSE 0.07), which a uniform reference-state offset could not produce, and an independent recomputation reproduces the closure output—so the mean offset is genuine decoder bias, not a units artifact. We headline the convention-independent bound (§6.2) as the load-bearing claim. (ii) The assumption-free constant-offset margin is convention-specific (full COSMO-SAC) and *inverts* under the deployed residual-only default; there the separation rests on the convention-independent bound  $B_{\text{closure}} \geq \text{MSE} - B_{\text{insuff}}^{\text{up}}$  and the more-physics-worse effect (which we verified is robust to the molecular-volume basis). (iii) The measurement is at infinite dilution and 298 K on the VT-2005-matchable, low- $\gamma$  subset; the non-ideal amide-solvent regime is untested. (iv) Absolute scaffold  $\ln x_2$  accuracy is encoder-bound; it sits well *above* the inter-lab replicate-reproducibility floor (0.15–0.31 in  $\ln x_2$ ; §6.8), so label noise does not explain the ceiling, but it is near the practical difficulty of new-solute extrapolation set by the current state of the art. (v) The compensating-surrogate result rests on a single converged one-model isolation control ( $n=44$  matched molecules; §6.3): the reported 53% departure, its low-rank structure (top-two modes 73%), and its  $3.3\times$  out-of-sample transfer supersede an earlier short pilot that reached only  $\sim 16\%/\sim 1.7\times$  before convergence. The remaining hedges are the sample size and that the clean isolation is a single GPU run. This paper is not an accuracy claim.

## 8 Conclusion

Feeding a physics-informed solubility model its true physical inputs made it worse. What the paper contributes is a way to measure that failure: with an externally grounded latent, the true-input excess risk splits into a decoder-misspecification term and an input-insufficiency term, and on the infinite-dilution, low- $\gamma$  subset we can measure, the ceiling is most likely the closure rather than the inputs ( $P \approx 0.78$  under two-way solute $\times$ solvent clustering, 0.89 single-axis; not certified at conventional confidence; label noise unestimable and assumed zero). The practical implication is to fix the map itself, since within this closure truer inputs surface its bias and an output-space

## When a physics prior helps vs. hurts

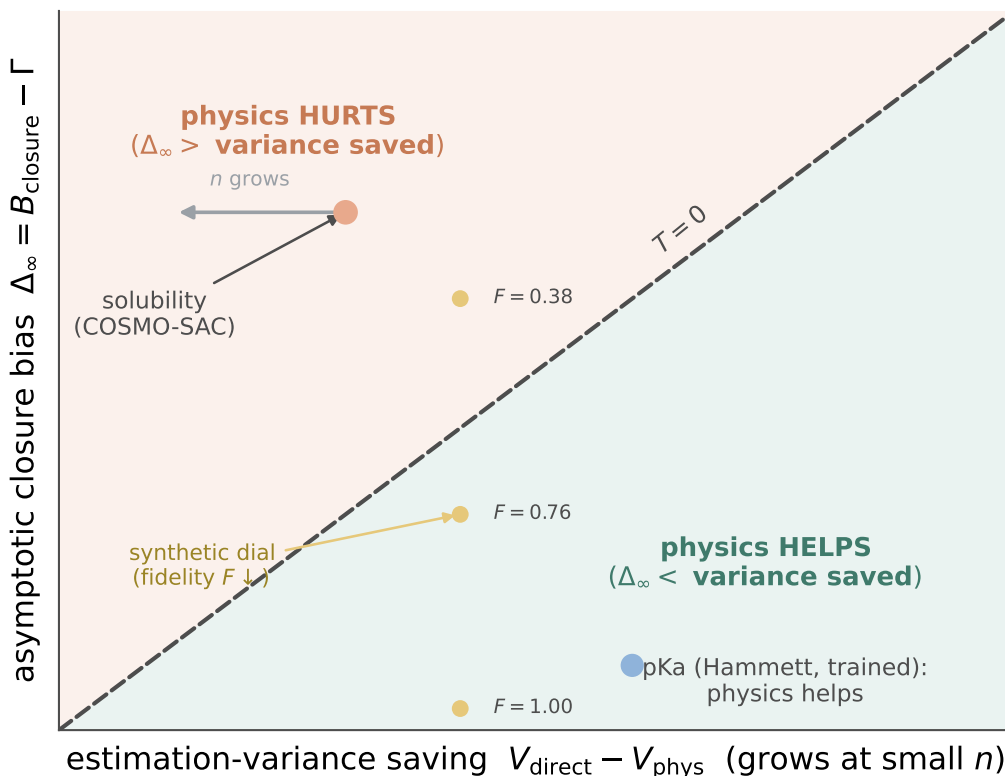


Figure 15: When a physics prior helps versus hurts. The physics tax has sign  $\mathcal{T} = \Delta_\infty -$  (variance saved): a fixed-closure prior helps only below the  $\mathcal{T} = 0$  diagonal, where the estimation variance it saves exceeds its asymptotic closure bias  $\Delta_\infty$  (the large- $n$  asymptote;  $= B_{\text{closure}} - \Gamma$  under A2). The variance saving shrinks as  $n$  grows, so systems drift left: solubility/COSMO-SAC sits deep in the “hurts” region and moves further in with more data, pKa through a Hammett closure lands on the “physics helps” side as a *trained* comparison (§6.6)—though a stratified check shows the help is not cleanly a closure-fidelity effect—and the synthetic dial climbs out of “helps” as closure fidelity  $F$  falls. Conceptual—only the sign is measured, not the boundary’s exact shape.

correction is inert. Neither failure is universal: raising closure fidelity (COSMO-SAC-2010/dsp dissolves the certificate) or supervising the latent toward truth moves the system out of the hurts-corner, so whether grounding helps is governed by two axes—closure fidelity and how the latent is trained (Fig. 14). The same grounding also indicates, where the closure binds, that the learned  $\sigma$ -profile is a closure-compensating surrogate rather than the physical quantity—an interpretability cost of physics-informed bottlenecks that the external latent makes measurable: a 53%, low-rank, transferable departure from truth on a converged one-model control.

## Data and code availability

All code and the intermediate result artifacts needed to regenerate every figure and table are available at [\[repository URL\]](#) (commit [\[hash\]](#)), released under [\[license\]](#). The solubility corpus is BigSolDB 2.0 [13]; the true  $\sigma$ -profiles are the VT-2005 database, ingested to `results/sigma_profile_artifact/si`; the experimental infinite-dilution activity coefficients are compiled from ThermoML (`notebooks/data/raw/idac.c`).

The Bemis–Murcko scaffold split is produced by `scripts/prepare_data.py`. The controlled comparison (Fig. 6, §F.3) is produced by `scripts/experiments/run_e5_sigma_grounding.sh` and aggregated by `scripts/analysis/run_e5_comparison.py` over `results/e5_sigma_grounding/seed_{42,43,44}/`; the 3-seed aggregate is tabulated in `results/e5_sigma_grounding/THREE_SEED_SUMMARY.md`. The closure–insufficiency decomposition (Table 3, Figs 7 and 8) is regenerated deterministically by `scripts/analysis/run_b_insuff_decomposition.py`  $\rightarrow$  `results/b_insuff/decomposition.json` from the committed matched-pair table `results/b_insuff/matched_pairs.csv` (the  $n=60$  VT-2005-matchable pairs). The second-domain pKa measurement (§6.6) uses the OPERA experimental compilation [18]: `scripts/data/build_pk_a_hammett_subset.py` assigns the oracle Hammett  $\sigma$  and `scripts/analysis/run_pk_a_real_decomposition.py` produces the decomposition ( $\rightarrow$  `results/pk_a_hammett/real_decomposition.json`) with the controlled semi-synthetic validation in `scripts/analysis/run_pk_a_hammett_probe.py` and the trained encoder-vs-DirectGNN comparison in `scripts/experiments/run_pk_a_trained_comparison.py`. The synthetic fidelity dial (Fig. 11) is `scripts/analysis/run_fidelity_dial.py`, and the conceptual schematics (Figs 3, 5, 4, 15) and the architecture figure (Fig. 2) are `scripts/analysis/make_concept_figures.py`. The model checkpoints and the full per-arm prediction CSVs are large and are deposited at [Zenodo DOI] rather than in git; the named scripts regenerate them from the raw sources. The manuscript compiles with `make` in `paper/`.

## Supplementary Information

### A Supplementary methods

**COSMO-SAC constants.** The differentiable COSMO-SAC layer uses the standard VT-2005 / Lin–Sandler constants: effective segment area  $a_{\text{eff}} = 7.5 \text{ \AA}^2$ , electrostatic-misfit constant  $\alpha' = 16466.72 \text{ kcal \AA}^4 \text{ mol}^{-1} \text{ e}^{-2}$ , hydrogen-bond coefficient  $c_{\text{hb}} = 85580$  with cutoff  $\sigma_{\text{hb}} = 0.0084 \text{ e \AA}^{-2}$ , coordination number  $z = 10$ , and Staverman–Guggenheim normalisers  $r_0 = 66.69 \text{ \AA}^3$ ,  $q_0 = 79.53 \text{ \AA}^2$ ; the  $\sigma$ -profile grid is 51 bins on  $[-0.025, 0.025] \text{ e \AA}^{-2}$ . The segment-activity fixed point is damped ( $\lambda = 0.7$ ) and run for 16 iterations at train time and 30 at evaluation.

**Crystal head.**  $T_m = 100 + 600 \sigma(\cdot) \in [100, 700] \text{ K}$ ,  $\Delta H_{\text{fus}} = S_H \text{ softplus}(\cdot)$ ; the group-contribution  $T_m$  prior is affine- calibrated on the train split before entering the head.

**Loss.** Only the following terms are load-bearing for the claims: a bin-weighted Huber on  $\ln x_2$ ; auxiliary  $\sigma$ -profile (earth-mover), crystal  $(T_m, \Delta H_{\text{fus}})$ ,  $\ln \gamma_2^\infty$ , and Hansen losses; and van’t Hoff slope and monotonicity regularisers. The full  $\sim 30$ -term taxonomy, per-phase weights, graph featurisation, and encoder hyperparameters are in the training configs (`configs/physics_grounded.yaml` and `config.py`); every reported run pins its resolved config in `results/*/resolved_config.json`.

### B Proofs

*Proof of Lemma 1 (non-identifiability).* Write  $u = 1/T$  and  $h = \Delta H_{\text{fus}}/R$ ,  $u_m = 1/T_m$ . At infinite dilution and with  $\Delta C_p = 0$  the observable is

$$\mu(u) = -\Phi(u) - \ln \gamma_2^\infty(u) = -h(u - u_m) - (c + du) = -(h + d)u + (hu_m - c),$$

using  $\ln \gamma_2^\infty(u) = c + du$  (affine in  $u$  by hypothesis). Thus  $\mu$  depends on the parameters  $\theta = (\Delta H_{\text{fus}}, T_m, c, d)$  only through the pair  $(S, K) = (-(h + d), hu_m - c) \in \mathbb{R}^2$ . The activity block spans



this plane,  $\partial(S, K)/\partial c = (0, -1)$  and  $\partial(S, K)/\partial d = (-1, 0)$ ; hence for *any* crystal value  $(h, u_m)$  and any target  $(S, K)$  there is an activity pair  $d = -S - h$ ,  $c = hu_m - K$  reproducing it. The likelihood level set therefore contains the full two-dimensional family  $\{(h, u_m)\}$ , so  $(\Delta H_{\text{fus}}, T_m)$  are unidentified. Equivalently, the crystal score  $\partial(S, K)/\partial h = (-1, u_m)$  lies in the span of the activity columns, so profiling out  $(c, d)$  annihilates it and  $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = 0$ . At finite dilution  $\ln \gamma_2(u) = \ln \gamma_2^\infty(u) (1 - x_2)^2$  adds  $\ln \gamma_2^\infty(u) (2x_2 - x_2^2) = O(x_2)$ , which is not affine in  $u$  because  $x_2 = x_2(u)$ ; the crystal score then has a component outside the activity span of order  $x_2$ , so  $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = O(x_2^2) > 0$ .  $\square$

*Proof of Lemma 2 (class-dependent information).* This is the semiparametric efficient-score construction [29] (here finite-dimensional and Gaussian, so equivalently the partitioned-information / Frisch–Waugh–Lovell identity). With Gaussian noise of variance  $\sigma^2$ , parameter-of-interest score  $v = \partial\mu/\partial\Delta H_{\text{fus}}$ , and nuisance tangent space  $\mathcal{T}_{\mathcal{H}}$  (the  $L^2$ -closure of the span of nuisance scores), the efficient score is the residual  $\tilde{v} = v - \Pi_{\mathcal{H}}v$  and the efficient information is  $\mathcal{I}_{\text{eff}} = \sigma^{-2}\|\tilde{v}\|^2$ . Here  $v = \partial\mu/\partial\Delta H_{\text{fus}} = -(u - u_m)/R \in \text{span}\{1, u\}$ . If  $\mathcal{T}_{\mathcal{H}} \supseteq \text{span}\{1, u\}$  then  $\Pi_{\mathcal{H}}v = v$  and  $\mathcal{I}_{\text{eff}} = 0$ , recovering Lemma 1; if  $\mathcal{T}_{\mathcal{H}}$  is a proper subspace not containing  $v$  (e.g. a fixed slope,  $\mathcal{T}_{\mathcal{H}} = \text{span}\{1\}$ ) then  $\Pi_{\mathcal{H}}v \neq v$  and  $\mathcal{I}_{\text{eff}} > 0$ , with a finite Cramér–Rao bound.  $\square$

*Proof of Lemma 3 (decomposition and bounds).* (a) *Identity.* Split the residual as  $m - g(z^*) = [m - \mathbb{E}(m \mid z^*)] + [\mathbb{E}(m \mid z^*) - g(z^*)]$ , square, and take expectations. The squared terms give  $\mathbb{E}[\text{Var}(m \mid z^*)] = B_{\text{insuff}}$  and  $\mathbb{E}[(\mathbb{E}(m \mid z^*) - g(z^*))^2] = B_{\text{closure}}$ . The cross term vanishes: since  $\mathbb{E}(m \mid z^*) - g(z^*)$  is  $\sigma(z^*)$ -measurable, the tower rule gives  $\mathbb{E}[(m - \mathbb{E}(m \mid z^*))(\mathbb{E}(m \mid z^*) - g(z^*))] = \mathbb{E}[(\mathbb{E}(m \mid z^*) - g(z^*))\mathbb{E}(m - \mathbb{E}(m \mid z^*) \mid z^*)] = 0$ . (b) *Jensen lower bound.* With  $\Delta(z^*) = \mathbb{E}(m \mid z^*) - g(z^*)$ ,  $B_{\text{closure}} = \mathbb{E}[\Delta^2] \geq (\mathbb{E}[\Delta])^2 = (\mathbb{E}[m] - \mathbb{E}[g])^2$  by Jensen, using  $\mathbb{E}[\mathbb{E}(m \mid z^*)] = \mathbb{E}[m]$ ; it requires only that  $m$  and  $g$  share a reference state, so that  $\mathbb{E}[m] - \mathbb{E}[g]$  is a genuine mean discrepancy. (c) *Law-of-total-variance upper bound.* For any binning  $B = \text{bin}(g(z^*))$ ,  $\sigma(B) \subseteq \sigma(z^*)$ , so conditioning on the coarser  $\sigma$ -algebra cannot reduce expected conditional variance:  $\mathbb{E}[\text{Var}(m \mid B)] \geq \mathbb{E}[\text{Var}(m \mid z^*)] = B_{\text{insuff}}$ . (d) *Convention-independence.*  $B_{\text{insuff}} = \mathbb{E}[\text{Var}(m \mid z^*)]$  is a functional of the law of  $(m, z^*)$  alone and does not involve  $g$ ; hence it is identical across closure conventions, and any  $\text{MSE}(g_1) - \text{MSE}(g_2)$  is entirely  $B_{\text{closure}}(g_1) - B_{\text{closure}}(g_2)$ .  $\square$

*Proof of Theorem 1 (grounding-gap sandwich).* Throughout assume  $m, g(z^*) \in L^2$  and  $g$ , every  $h \in \mathcal{E}$  measurable. (a)  $\Gamma \geq 0$ . By A1,  $\phi^* \in \mathcal{E}$  is a feasible encoder with  $g(\phi^*(x)) = g(z^*)$ , so  $R_{\text{e2e}} = \inf_h \mathbb{E}[(m - g(h(x)))^2] \leq \mathbb{E}[(m - g(z^*))^2] = R_{\text{orc}}$ ; the infimum need not be attained. Hence  $\Gamma = R_{\text{orc}} - R_{\text{e2e}} \geq 0$ . (b)  $R_{\text{e2e}} \geq R_{\text{free}}$ . Every  $g(h(x))$  is  $\sigma(x)$ -measurable, and  $R_{\text{free}} = \mathbb{E}[(m - \mathbb{E}(m \mid x))^2]$  is the infimum of  $\mathbb{E}[(m - p)^2]$  over *all*  $\sigma(x)$ -measurable  $p$ ; an infimum over the subclass  $\{g(h(\cdot)) : h \in \mathcal{E}\}$  is no smaller. (c)  $R_{\text{free}} = B_{\text{insuff}}$  under A2. From  $\text{Var}(m) = \mathbb{E}[\text{Var}(m \mid x)] + \text{Var}(\mathbb{E}(m \mid x)) = \mathbb{E}[\text{Var}(m \mid z^*)] + \text{Var}(\mathbb{E}(m \mid z^*))$  and A2 ( $\mathbb{E}(m \mid x) = \mathbb{E}(m \mid z^*)$ , so the two  $\text{Var}(\mathbb{E}(\cdot))$  terms agree), the two expected conditional variances agree:  $R_{\text{free}} = \mathbb{E}[\text{Var}(m \mid x)] = B_{\text{insuff}}$ . (d) *Sandwich and identity.* By Lemma 3,  $R_{\text{orc}} = B_{\text{insuff}} + B_{\text{closure}}$ . With (c),  $\Gamma = R_{\text{orc}} - R_{\text{e2e}} = B_{\text{closure}} - (R_{\text{e2e}} - R_{\text{free}})$ , and (b) gives  $R_{\text{e2e}} - R_{\text{free}} \geq 0$ , so  $0 \leq \Gamma \leq B_{\text{closure}}$ ; equality  $\Gamma = B_{\text{closure}}$  holds iff  $R_{\text{e2e}} = R_{\text{free}}$ . Finally  $B_{\text{closure}} = 0$  forces  $0 \leq \Gamma \leq 0$ , i.e.  $\Gamma = 0$ ; contrapositively, under A1–A2,  $\Gamma > 0 \Rightarrow B_{\text{closure}} > 0$ . (e) *A2 is necessary for the certificate.* If A2 fails, take  $x \in \{a, b\}$  equiprobable,  $z^* \equiv 0$  (so A1 holds),  $m(a) = +1, m(b) = -1$ , and  $g$  with  $g(0) = 0 = \mathbb{E}(m \mid z^*)$  so  $B_{\text{closure}} = 0$ ; an encoder  $h^\dagger(a) = 1, h^\dagger(b) = -1$  with  $g(\pm 1) = \pm 1$  gives  $R_{\text{e2e}} = 0$  while  $R_{\text{orc}} = 1$ , hence  $\Gamma = 1 > 0 = B_{\text{closure}}$ . So  $\Gamma > 0$  alone does not certify  $B_{\text{closure}} > 0$ ; the decomposition (Lemma 3) does.  $\square$

*Proof of Proposition 1 (sign of the physics tax).* Write  $D(n) := V_{\text{direct}}(n) - V_{\text{phys}}(n)$ , so  $\mathcal{T}(n) = \Delta_{\infty} - D(n)$  with  $\Delta_{\infty} = R_{\text{e2e}} - R_{\text{direct},\infty}$ . (i)  $\mathcal{T}(n) < 0 \iff D(n) > \Delta_{\infty}$ , immediately. (ii) If  $\Delta_{\infty} > 0$  and  $D(n) \rightarrow 0$ , choose  $n^*$  with  $|D(n)| < \Delta_{\infty}$  for all  $n > n^*$  (possible since  $D(n) \rightarrow 0$ ); then  $\mathcal{T}(n) = \Delta_{\infty} - D(n) > 0$ . Monotonicity of  $D$  is *not* required for existence of  $n^*$ ; it is needed only for the stronger claim “ $\mathcal{T}(n) > 0$  at every  $n$ ,” which we treat as an empirical observation. (iii) Since  $V_{\text{phys}}(n) \geq 0$ ,  $D(n) \leq V_{\text{direct}}(n)$ : the variance a prior can save is bounded by the control’s own estimation variance. As the black box approaches the aleatoric floor  $V_{\text{direct}}(n) \rightarrow 0$ , so  $\limsup_n D(n) \leq 0$ ; with both estimators consistent ( $V_{\text{phys}}, V_{\text{direct}} \rightarrow 0$ ) one has  $D(n) \rightarrow 0$  and  $\mathcal{T}(n) \rightarrow \Delta_{\infty} \geq 0$ . When the control is asymptotically Bayes ( $R_{\text{direct},\infty} = R_{\text{free}}$ ) and A2 holds,  $\Delta_{\infty} = R_{\text{e2e}} - R_{\text{free}} = B_{\text{closure}} - \Gamma$  by (3).  $\square$

## C Per-arm tables and decomposition robustness

Table 6 gives the exact per-arm metrics behind Fig. 6: three seeds (mean $\pm$ sd) on the cross-arm intersection lock ( $n=5608$ , rows supervised and finite in every arm). Both closure families—NRTL and COSMO-SAC over a learned  $\sigma$ -profile—trail the DirectGNN control (the physics tax); the  $\sigma$ -oracle (true VT-2005 profile) is the worst arm, and restoring the Staverman–Guggenheim combinatorial term makes it slightly worse still—the signature of a misspecified map (§6.2).

Table 6: Per-arm scaffold-split metrics (3 seeds,  $n=5608$  cross-arm lock). Ordered by MAE; the physics closures trail DirectGNN and the true- $\sigma$  oracle is worst. Regenerated by `run_e5_comparison.py` over `results/e5_sigma_grounding/seed_{42,43,44}/`.

| Arm                                          | MAE             | $R^2$            |
|----------------------------------------------|-----------------|------------------|
| DirectGNN (no closure)                       | $1.70 \pm 0.03$ | $+0.42 \pm 0.03$ |
| NRTL closure                                 | $1.80 \pm 0.07$ | $+0.34 \pm 0.03$ |
| COSMO-SAC, learned $\sigma$ (grounded)       | $1.85 \pm 0.05$ | $+0.33 \pm 0.05$ |
| + Staverman–Guggenheim combinatorial         | $1.88 \pm 0.09$ | $+0.32 \pm 0.05$ |
| COSMO-SAC, ungrounded $\sigma$               | $2.04 \pm 0.04$ | $+0.19 \pm 0.01$ |
| COSMO-SAC, true $\sigma$ ( $\sigma$ -oracle) | $2.25 \pm 0.02$ | $-0.03 \pm 0.04$ |

**Robustness of the closure/insufficiency separation.** We stress the deployed-convention verdict  $B_{\text{closure}} > B_{\text{insuff}}$  (§6.2) three ways on the  $n=60$  set (`run_b_insuff_robustness.py`). (i) *Estimator sensitivity.* The  $B_{\text{insuff}}$  upper bounds are 0.40 ( $k$ -NN), 0.56 (binning/LOTV), 0.62 (random forest), and 0.88 (ridge)—computed with this section’s estimator settings, so they differ slightly from Table 3, a hyperparameter effect, not a convention one ( $B_{\text{insuff}}$  is convention-independent, not estimator-independent). The spread is a bias–variance ordering of the same target  $\mathbb{E}[\text{Var}(m \mid z)]$ : the local  $k$ -NN estimate is the least smoothed and sits lowest, while the global ridge smoother is the most regularized and sits highest. Because a *smaller*  $B_{\text{insuff}}^{\text{up}}$  is the *less* conservative input to  $B_{\text{closure}} \geq \text{MSE} - B_{\text{insuff}}^{\text{up}}$ , we do not lean on the  $k$ -NN value despite its being the smallest—it is the most optimistic, not the most trustworthy. With  $\text{MSE}_{\text{res}} = 1.47$  the verdict  $B_{\text{closure}} > B_{\text{insuff}}$  holds under the three lower estimators ( $k$ -NN, binning/LOTV, random forest) and fails only under the most conservative ridge estimator—non-certifying there, not refuting, and consistent with the clustered-bootstrap “likely but not certified” reading below. (ii) *Leave-one-solvent-out.* Dropping each of the 17 solvents in turn, the verdict holds in *all* 17 folds under the law-of-total-variance bound; under the random-forest estimator it holds in only 2/17, because a random forest over-fits

conditional variance in 102 dimensions at  $n=60$  (its out-of-fold MSE approaches  $\text{Var}(m)$ ), so its leave-one-out estimate is unstable—an estimator artifact, not a sign flip. (iii) *Label noise*. Injecting zero-mean Gaussian noise  $\sigma_\eta \in \{0.2, \dots, 0.8\}$  into  $m$  leaves the estimated  $B_{\text{closure}}$  lower bound stable ( $0.84 \rightarrow 0.70$ ), as expected since  $B_{\text{closure}} = \text{MSE} - \mathbb{E}[\text{Var}(m \mid z^*)]$  is invariant to zero-mean noise. The separation is thus robust under a principled insufficiency estimator and noise-stable, but its numerical strength is estimator-dependent. (iv) *Clustered-bootstrap interval*. A solvent-clustered bootstrap (3000 resamples of the 17 solvents, LOTV bound) makes the confidence explicit: the  $B_{\text{closure}}$  lower bound has median 0.93 ( $\text{CI}_{90}$  [0.40, 1.62]) and the certification margin  $B_{\text{closure}} - B_{\text{insuff}}$  has median 0.36 ( $\text{CI}_{90}$  [-0.22, +0.97]), with  $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.89$ . The separation is therefore *likely* but not certified at the 90% level—consistent with the earlier  $\approx 0.85$  figure (hedge 3)—which is why the main text treats it as supporting rather than decisive.

**In-regime ( $\gamma$ -stratified) separation.** The  $n=60$  measurement is matched at low  $\gamma$ , while the paradox spans all  $\gamma$ . Stratifying the  $n=60$  set by  $\ln \gamma_2^\infty$  (`run.b.insuff.regime.py`) tests whether closure-dominance is a corner artifact. It is not: the verdict  $B_{\text{closure}} > B_{\text{insuff}}$  (tightest of the binning/RF/ $k$ -NN upper bounds) holds in every stratum and *strengthens* toward higher  $\gamma$  (Table 7)—in the highest stratum ( $\overline{\ln \gamma_2^\infty} = 2.9$ , approaching the non-ideal regime)  $B_{\text{closure}} \geq 3.0$  dominates  $B_{\text{insuff}} \leq 0.3$  by an order of magnitude. We caution that the high- $\gamma$  strata are small ( $n = 17\text{--}30$  in 102 dimensions): the law-of-total-variance bound can be pushed toward zero by too-few-points-per-bin, which mechanically widens the margin exactly where data is thinnest, so it is the large ( $\sim 10\times$ ) separations rather than the precise values that carry the qualitative reading. This supports—but does not prove—the conjecture that the same closure misspecification drives the all- $\gamma$  paradox.

Table 7: Closure/insufficiency verdict across  $\ln \gamma_2^\infty$  strata of the matched set (deployed residual-only convention;  $B_{\text{insuff}}^{\text{up}}$  is the tightest of the binning / RF /  $k$ -NN upper bounds). The separation holds in every stratum and grows with  $\gamma$ .

| Stratum                                 | $n$ | $\overline{\ln \gamma_2^\infty}$ | $B_{\text{insuff}} \leq$ | $B_{\text{closure}} \geq$ |
|-----------------------------------------|-----|----------------------------------|--------------------------|---------------------------|
| all                                     | 60  | +0.75                            | 0.40                     | 1.07                      |
| lower $\gamma$ ( $m \leq \text{med.}$ ) | 30  | -0.40                            | 0.07                     | 0.95                      |
| higher $\gamma$ ( $m > \text{med.}$ )   | 30  | +1.90                            | 0.28                     | 1.65                      |
| high $\gamma$ ( $m > 1.5$ )             | 17  | +2.94                            | 0.32                     | 3.04                      |

The closure/insufficiency decomposition (full/res conventions; law-of-total-variance (LOTV), random-forest (RF), ridge, and kNN estimators; solvent-clustered bootstrap; the pyridine stratum) is tabulated in §6.2 and regenerated by `run.b.insuff.decomposition.py`.

**The four hedges (full).** The adversarial audit could not invert the verdict but sharpened its scope:

1. *Convention*. The deployed default is residual-only. Under it the *constant-offset* (Jensen) lower bound inverts ( $B_{\text{closure}} \geq 0.43 < B_{\text{insuff}} \leq 0.62$ ), so that assumption-free bound does not separate, and the full-convention margin (the sharpened  $\text{MSE} - B_{\text{insuff}}$  bound, 1.27 vs 0.53) is a full-COSMO-SAC-only result. What separates under the deployed convention is the leakage-immune bound  $B_{\text{closure}} \geq \text{MSE}_{\text{res}} - B_{\text{insuff}}^{\text{LOTV}} \approx 0.85 > 0.62$ , on which we headline, together with the combinatorial Tier 2 fact.

2. *Confound*. The closure error is predominantly a *between-solvent* systematic bias ( $\sim 64\%$  of total error); within the largest single-solvent stratum (pyridine,  $n=20$ ) the closure is nearly correct (stratum MSE 0.07). Because solvent  $\sigma$  is part of  $z^*$ , these offsets are still decoder error and give an independent solvent-conditional lower bound  $B_{\text{closure}} \geq 0.89$  (full)/0.56 (res).
3. *Confidence*. Honest confidence on the full-convention constant-offset separation is the solvent-clustered bootstrap  $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.85$  (0.83 vs the random-forest upper bound, 0.89 vs the law-of-total-variance bound); we report the solvent-clustered scheme rather than the less conservative i.i.d.-pair one because solvent dependence dominates. The leakage-immune margin (0.85 vs 0.62) is a point estimate. The decomposition is checkpoint/seed-independent (the closure has zero learnable parameters).
4. *Selection/scope*. The 60 matchable pairs are the low-to-moderate  $\gamma$  corner (matched mean  $\ln \gamma_2^\infty = 0.75$  vs 3.56 for VT-2005-unmatchable pairs,  $t = -7.44$ ; the strongly non-ideal amide-solvent regime is untested because VT-2005 lacks those profiles). This narrows scope but is *conservative*, since the untested regime is a harder closure test. The target is  $\ln \gamma_2^\infty$  at 298 K, so the numeric ceiling is not claimed to transfer unchanged to the finite-composition solubility regime.

## D External baselines and ablations

We retrained the direct SOTA (FastSolv [1]) and the physics-informed SOTA (SolProp [30]) on our corpus and evaluated them on a by-solute test split (`test_solute.csv`,  $n \approx 10\text{--}12\text{k}$ ). In  $\ln x_2$  they reach MAE 1.94 (FastSolv,  $R^2$  0.23) and 2.34 (SolProp, calibrated,  $R^2 -0.18$ ); in  $\log_{10} S$ , 0.87 and 1.04. Neither beats our matched arms, and the physics-informed SolProp trails the direct FastSolv—the same ordering we find internally between the physics closures and DirectGNN. These runs use a different (by-solute) test file than the scaffold split of the headline comparison, so they bound the task’s difficulty rather than rank directly against the scaffold numbers of Table 6; a like-for-like scaffold evaluation of the external models is left to the deposited artifacts. The regime-level reading is that on this corpus no black box, ours or external, escapes the accuracy ceiling—consistent with the structural account of §7. Per-arm ablations (NRTL vs COSMO-SAC, combinatorial on/off) are in Table 6.

## E Supplementary mechanism results

**Non-identifiability, empirically (the corrupted twin).** Training a twin model on data whose crystal labels are corrupted (every  $T_m$  shifted by +273 K) leaves the *solubility* fit essentially unchanged— $\ln x_2$  MAE 1.835 versus 1.812 on the corrected data—while the recovered crystal grounding differs by hundreds of kelvin ( $T_m$  MAE 258 K versus 45 K). Two very different crystal/activity splits produce the same solubility, which is Lemma 1 made concrete.

**Crystal grounding on physical labels.** When the crystal head is supervised on measured melting points it predicts  $T_m$  to 45 K MAE, about  $4\times$  better than a Joback group-contribution reference—so the flat downstream result of §F.2 is a dose/identifiability effect, not an inability to learn crystal properties.

**The  $\Delta C_p = 0$  approximation.** The ideal term omits the constant-heat-capacity correction; a group-contribution audit finds a nonzero  $\Delta C_p$  for 1441 of 1525 solutes (108,287 rows), so this is a bounded model-form approximation. It cannot produce either headline result. The closure/insufficiency decomposition is measured on the activity target ( $\ln \gamma_2^\infty$  at 298 K), where  $\Delta C_p$  does not enter at all; and the grounding paradox is an arm-to-arm difference (learned- $\sigma$  vs. oracle- $\sigma$ ) whose two arms share the *identical* crystal term, so any  $\Delta C_p$  error cancels exactly in the comparison. It is thus orthogonal to the activity closure, not a source of the paradox.

**Fusion-supervision scarcity.** Direct  $\Delta H_{\text{fus}}$  labels are extremely sparse: 1279 training rows across 31 solutes, and essentially none in validation or test. This is the quantitative backbone of the identifiability argument—there is almost no signal with which to pin the crystal term from data.

**The NRTL branch trains (a trainability check).** The physics tax is not an under-training artifact of the activity branch. In the headline NRTL arm the head produces solvent-varying interaction parameters— $\tau_{12}$  over the  $n=8103$  test pairs has mean 0.97 and std 2.5 (range  $[-2.6, 22]$ ), far from collapsed—yet the arm still trails the closure-free DirectGNN (1.80 vs 1.70; Table 6). A trained, non-degenerate activity branch beaten by the control is evidence that the tax is structural, not a dead branch. (The core closure-misspecification result—the  $\sigma$ -oracle and the decomposition—is in any case training-*independent*: it feeds the true VT-2005 profiles through the fixed closure with no encoder optimization, so it cannot be a training pathology.)

**Structured temperature extrapolation.** Withholding *temperature* rather than structure (train at  $T \leq 309.75$  K, test at higher  $T$ ;  $n = 3343$ ), the affine-in- $1/T$  hypothesis classes extrapolate far better than an unstructured descriptor regressor (Table 8); the van’t Hoff class also predicts the *direction* of the high- $T$  shift correctly 99% of the time versus 40% for the random forest. This is a property of the structured hypothesis class—the trained neural models are weaker than the van’t Hoff baseline—not an accuracy claim for our model.

Table 8: Same-pair high-temperature extrapolation ( $n = 3343$ , train  $T \leq 309.75$  K). Structured  $1/T$  classes extrapolate; a descriptor regressor and constant baselines do not. “dir.” is the fraction of pairs whose high- $T$  shift direction is predicted correctly.

| Model                         | MAE  | $R^2$ | dir. |
|-------------------------------|------|-------|------|
| per-pair van’t Hoff ( $1/T$ ) | 0.37 | 0.89  | 0.99 |
| per-pair linear in $T$        | 0.41 | 0.85  | 0.99 |
| per-pair last-low- $T$ hold   | 1.20 | 0.69  | 0.01 |
| RF (Morgan + $T$ )            | 1.29 | 0.66  | 0.40 |
| per-pair mean                 | 1.46 | 0.56  | 0.01 |

## F Supplementary diagnostics and broad negatives

### F.1 Identifiability and compensation

Lemma 1 asserts that the crystal/activity split is not recoverable from solubility data alone. Two diagnostics make this concrete. The first is a numerical Fisher audit of the SLE design, computed on synthetic data and independent of any particular activity closure. We evaluate the Fisher

Table 9: Fisher audit of the SLE design at  $\Delta T = 40$  K (synthetic, closure-independent). Solubility alone leaves the crystal parameter near-unidentified; external single-component labels close the deficiency.

| Supervision regime              | cond( $I$ )       | CRLB sd( $\Delta H_{\text{fus}}$ ) [J/mol] |
|---------------------------------|-------------------|--------------------------------------------|
| Solubility only                 | $2.3 \times 10^5$ | 19,895                                     |
| + $T_m$ label                   | $5.4 \times 10^3$ | 3,151                                      |
| + $T_m + \Delta H_{\text{fus}}$ | $1.6 \times 10^3$ | 1,689                                      |

information matrix  $I$  of the observation model at a reference operating point and read off the conditioning  $\text{cond}(I)$  and the Cramér–Rao lower bound on the standard deviation of  $\Delta H_{\text{fus}}$  under three supervision regimes. Table 9 reports the result at a temperature window of  $\Delta T = 40$  K. With solubility measurements only, the design is severely ill-conditioned and the CRLB on  $\Delta H_{\text{fus}}$  is roughly 20,000 J/mol, comparable to the quantity being estimated. Adding a direct  $T_m$  label drops the conditioning by nearly two orders of magnitude and cuts the bound to about 3,200 J/mol; supplying  $T_m$  and  $\Delta H_{\text{fus}}$  together brings it to about 1,700 J/mol. The ill-conditioning is a property of the design rather than of noise level, and it is closed only by external single-component supervision, which is the mechanism the aux-stream pattern supplies.

Widening or narrowing the temperature window does not rescue the solubility-only case. A  $\Delta T$  sweep moves the solubility-only conditioning from  $1.7 \times 10^3$  at a wide window to  $2.3 \times 10^5$  at 40 K and to  $1.9 \times 10^{16}$  at 10 K, so multi-temperature data alone does not resolve the split; the two parameters remain confounded along the ideal-term direction. Restricting the activity slope rather than adding labels helps, but only partially: it cuts  $\text{sd}(\Delta H_{\text{fus}})$  to 8,251 J/mol, still well above the label-supervised regimes. A second, model-side diagnostic must be read with more care, and we keep it out of the load-bearing argument. On the supervised test set, with a group-contribution (Joback) crystal reference, the crystal and activity residuals are strongly anti-correlated,  $\text{corr}(\delta\Phi, \delta \ln \gamma_2) = -0.962$ . This is an *accounting artifact*, not evidence of a learned split: the two residuals sum to the (negative)  $\ln x_2$  fit residual, so their correlation tends to  $-1$  as the fit improves, and a permutation null that destroys any learned coupling still reproduces most of it ( $-0.90$  to  $-0.93$ ); the model’s own factors barely co-vary ( $\text{corr}(\hat{\Phi}, \ln \gamma_2) = -0.19$ ). The large apparent crystal offset ( $\delta\Phi \approx -7.1$ ) is likewise not a usable mis-grounding metric — it is dominated by reference error, since the Joback reference is physically implausible on these drug-like polycyclics (a substantial fraction of predicted  $T_m$  exceeds 1000 K), and capping it at a physical bound collapses the offset to  $\approx -1.5$ . We therefore rest the non-identifiability claim on the reference-free Fisher/CRLB audit above, and relegate the compensation figures to the SI (§E) as a cautionary diagnostic rather than a decomposition-quality measurement.

## F.2 Grounding the crystal factor: a dose-limited null

A natural test of whether the ceiling is inputs rather than closure is to ground a *different* factor—the crystal term—on abundant external data. We trained the identical model without and with an external single-component crystal-property pool of roughly 15,000 melting points, added through the auxiliary-stream mechanism after excluding any pool solute whose Bemis–Murcko scaffold appears in the validation or test split. The pool did not improve prediction of the crystal properties it supplies (melting-point MAE  $45.0 \rightarrow 46.6$  K, effectively unchanged, if anything marginally worse), and downstream solubility did not improve either ( $\ln x_2$  MAE  $1.81 \rightarrow 1.88$ ; two nominally favorable but slight shifts, in  $\ln x_2$   $R^2$  and the crystal offset  $|\delta\Phi|$ , do not amount to a gain). We flag this

as *dose-limited and inconclusive rather than a clean null*. In this run the crystal auxiliary stream was homeopathically dosed—a small fraction of optimizer steps at low loss weight, so the model saw the external pool less than one full pass at the selected checkpoint—and a stream that did not even lower the MAE of the property it directly supervises cannot license a strong conclusion. What it shows is that *this dilute intervention* does not move the ceiling, not that external crystal labels cannot. We therefore do *not* read it through the activity-path  $B_{\text{closure}}/B_{\text{insuff}}$  decomposition: the crystal “closure” is the near-exact ideal-solubility term  $\Phi = (\Delta H_{\text{fus}}/R)(1/T - 1/T_m)$ , not a misspecified map, so the flat  $T_m$  recovery is better read as the crystal/activity split’s structural non-identifiability (Lemma 1) and the encoder’s transfer limitation (§6.8) than as closure dominance. A properly dosed crystal-grounding test is left to future work.

### F.3 Chemistry-resolved structure of the physics tax

Across the full test set the two arms are separated by little more than label noise. On the 3-seed scaffold split the DirectGNN control reaches an  $\ln x_2$  MAE of  $1.70 \pm 0.03$ , while the grounded physics arm (learned- $\sigma$  COSMO-SAC closure) reaches  $1.85 \pm 0.05$ , for a global  $\Delta\text{MAE} = \text{MAE}(\text{physics}) - \text{MAE}(\text{control}) = +0.14$  (positive meaning the closure adds cost; the per-seed values span  $+0.05$  to  $+0.25$ , so this point estimate is itself only a 3-seed mean, not a tight one; it is computed on unrounded per-seed means, so the rounded  $1.70/1.85$  above differ by  $0.15$ ). At this granularity the physics bottleneck is a small, near-noise tax on accuracy. The interesting structure appears only once the error is resolved along chemical axes, where the average tax turns out to be an average over regimes that disagree in sign.

The solvent axis is the sharp one. Binning test pairs by solvent class and recomputing  $\Delta\text{MAE}$  per seed (Table 10) shows the cost concentrated in solvents that hydrogen-bond strongly and directionally, precisely the regime in which the COSMO-SAC closure is least faithful and  $|\ln \gamma_2|$  is largest. Water carries the heaviest penalty, and the effect decreases monotonically as the donor strength of the class softens. One class inverts the sign: amide solvents, which are aprotic dipolar hydrogen-bond acceptors and therefore closer to the closure’s well-specified regime, are the seed-robust stratum in which the tax reverses sign ( $\Delta\text{MAE} = -0.22 \pm 0.11$  over seeds). We flag this as exploratory rather than a confirmed win: across a family of nine simultaneous solvent-class comparisons the amide interval is an uncorrected per-class pair bootstrap, and the three-seed spread alone does not exclude zero; the halogenated class shows a nominal improvement of similar magnitude but is not seed-robust. The robust reading is the *direction* — the physics tax shrinks and can turn negative as solvent hydrogen-bond donation weakens toward the acceptor-dominated regime the closure was built for — not the significance of any single class. This solvent-resolved pattern is the chemistry-resolved shadow of the same synthetic dial we used to vary closure fidelity in controlled experiments.

The solute axis tells a compatible but weaker story (Table 11). The closure costs most on ordinary neutral organics—oxygenated solutes, polyaromatics, and heterocycles—while an earlier hypothesis, that it should hurt most on charged, high- $|\ln \gamma_2|$  solutes such as salts, did not replicate once the split was corrected: the charged/salt stratum is neutral. The clean penalty falls on neutral organics, not on the ionic tail we had expected to be hardest.

The most suggestive pattern is a novelty inversion that runs opposite to the usual coverage intuition: the physics arm is neutral on the most novel chemistry (nearest-train Tanimoto  $\leq 0.3$ ) and hurts most on the most familiar ( $> 0.8$ ; Table 11). We flag this as exploratory, not a confirmed effect: the seed spread is not a within-bin sampling interval over which solutes fall in the stratum, and the Tanimoto binning is one of several strata families we examine (solvent class, solute class, similarity) without multiplicity control. Read as a direction rather than a tested quantity, it

Table 10: Per-solvent-class physics tax  $\Delta\text{MAE} = \text{MAE}(\text{physics}) - \text{MAE}(\text{control})$  on the scaffold split (mean  $\pm$  sd over 3 seeds; positive means the closure adds cost). The intervals are uncorrected per-class pair bootstraps across a family of nine simultaneous comparisons; the two negative (physics-helps) entries are exploratory, not multiplicity-controlled.

| Solvent class   | $\Delta\text{MAE}$ (mean $\pm$ sd) |
|-----------------|------------------------------------|
| Water           | $+0.52 \pm 0.35$                   |
| Carboxylic acid | $+0.39 \pm 0.16$                   |
| Sulfoxide       | $+0.26 \pm 0.24$                   |
| Nitrile         | $+0.21 \pm 0.19$                   |
| Alcohol         | $+0.15 \pm 0.14$                   |
| Aromatic        | $+0.03 \pm 0.63$                   |
| Hydrocarbon     | $+0.02 \pm 0.50$                   |
| Halogenated     | $-0.29 \pm 0.36$                   |
| Amide           | $-0.22 \pm 0.11$                   |

Table 11: Per-solute-class and novelty-stratum physics tax  $\Delta\text{MAE} = \text{MAE}(\text{physics}) - \text{MAE}(\text{control})$  on the scaffold split (positive means the closure adds cost). Solute-class entries are point estimates; the Tanimoto strata carry the 3-seed spread. Exploratory: no multiplicity control across the strata families (§F.3).

| Solute stratum              | $\Delta\text{MAE}$ |
|-----------------------------|--------------------|
| Oxygenated                  | $+0.32$            |
| Polyaromatic                | $+0.27$            |
| Heterocycle                 | $+0.17$            |
| Halogenated-aromatic        | $\approx 0$        |
| Charged / salt              | $-0.04 \pm 0.19$   |
| Tanimoto $\leq 0.3$ (novel) | $-0.04 \pm 0.18$   |
| Tanimoto $> 0.8$ (familiar) | $+0.51 \pm 0.09$   |

suggests the ceiling on the physics path is not coverage or extrapolation but the closure’s systematic misfit—masked on novel solutes by the large errors both arms already incur there, and exposed on easy, well-covered solutes where the direct control is accurate enough for a structural bias in the closure to become the dominant residual—consistent with the same closure-misspecification reading as the rest of the paper. A multiplicity-controlled, within-bin-bootstrapped test is left to future work. Read together, the maps say the physics decomposition pays where the closure is misspecified relative to the solvent chemistry, and it can only turn into a win in the narrow acceptor-dominated regime the closure was built to describe.

## G The pKa closure: full construction, measured sweep, and trained comparison

This appendix gives the full second-closure instantiation summarized in §6.6: a fixed Hammett linear-free-energy relationship (LFER)  $g(\sigma) = \text{p}K_{a,0} - \rho\sigma$  over the Hammett substituent constant  $\sigma$  (with  $\text{p}K_{a,0}, \rho$  the per-scaffold Hansch–Leo–Taft reaction-series constants), for which Hansch–



Leo tabulated constants provide an external oracle  $z^*$ . The analogy is tight: as with the  $\sigma$ -profile through COSMO-SAC, a learnable physical descriptor is pushed through a fixed, approximate, differentiable map. The two classic LFER failure modes fall cleanly onto the two error channels of Lemma 3: *resonance saturation*—a curvature in  $\sigma$  that the linear closure structurally cannot represent—is a mis-map of a function of  $z^*$  and lands in  $B_{\text{closure}}$ ; *ortho/steric* field effects, absent from the tabulated constants, are information the intermediate lacks and land in  $B_{\text{insuff}}$ . The chemical scope is a fidelity sweep: meta/para monosubstituted benzoic acids, phenols, and anilinium ions are the near-exact ( $B_{\text{closure}} \approx 0$ ) regime, ortho (insufficiency-dominated) and polyfunctional the low-fidelity one.

This decomposition is directly visible before any training. Pushed through the fixed Hammett closure, *p*-nitrophenol reads  $\text{p}K_a \approx 8.3$  against an experimental  $\approx 7.1$  (and *p*-nitroaniline overshoots similarly): the linear closure is off by  $\sim 1$  unit exactly where through-conjugation ( $\sigma^-$ ) breaks it— $B_{\text{closure}}$  in the wild.

A controlled semi-synthetic construction on real tabulated  $\sigma$  (known ground-truth channels) validates the estimator and reproduces the paper’s central distinction in this second closure (Fig. 16). Sweeping closure fidelity with a sufficient intermediate, the grounding gap  $\Gamma$  tracks  $B_{\text{closure}}$  (the plug-in estimate recovers the truth to within a few percent, e.g. 0.20 vs 0.21, 0.81 vs 0.81), the case where Theorem 1’s proxy is valid. Sweeping the ortho/steric channel with a *well-specified* closure gives  $B_{\text{closure}} = 0$  throughout yet  $\Gamma > 0$  and growing—the  $\text{p}K_a$  realization of the conflation that makes  $\Gamma$  a symptom rather than a certificate. The oracle- $\sigma$  pipeline (scaffold-resolved substituent  $\rightarrow \sigma$  assignment) and this validation extend directly to real data. On the OPERA experimental  $\text{p}K_a$  compilation [18] (6,502 compounds), applying the same decomposition to the fixed Hammett closure yields a *measured* fidelity sweep: on the clean meta/para series ( $n=161$ ) the closure predicts experimental  $\text{p}K_a$  to RMSE 0.63 with  $B_{\text{closure}}$  below the input-insufficiency floor (per-scaffold systematic offsets  $\leq 0.25$ , so the LFER is well specified there), while the ortho/heteroaromatic regime ( $n=1008$ ) degrades to RMSE 2.4, dominated by input insufficiency ( $B_{\text{insuff}} \approx 4.7$ —the tabulated  $\sigma$  carries no ortho information, exactly the a-priori channel assignment) with an additional closure misspecification  $B_{\text{closure}} \geq 1.1$  on top. This is the near-exact-to-degraded transition the closure is meant to show, now measured rather than asserted. This places the  $\text{p}K_a$  closure near the phase-diagram boundary in its clean regime. We close the loop with a trained comparison. On a Bemis–Murcko scaffold split of the Hammett-amenable set (1169 compounds, three seeds), a shared GIN encoder feeding the fixed LFER (a predicted  $\sigma$ ) beats the matched DirectGNN control overall (1.63 vs 1.84  $\text{p}K_a$  MAE,  $\Delta = -0.21$ ), placing  $\text{p}K_a$  on the “helps” side of the phase picture (Fig. 15). We read this cautiously, *not* as a clean confirmation of the sign rule through closure fidelity. Stratified, the help concentrates in the ortho/heteroaromatic regime ( $\Delta = -0.53$ ), not the clean meta/para one ( $\Delta = -0.10$ , within seed noise); and on the clean regime the fixed LFER on the *true*  $\sigma$  is excellent (oracle MAE 0.60) yet the trained arm is not (1.61), so the closure is well specified there but the small-data encoder cannot learn  $\sigma$  well enough to exploit it. Where the physics arm does win—ortho, where the closure is *misspecified* for the true  $\sigma$ —the *freely* learned  $\sigma$  can absorb ortho information the tabulated Hammett constant lacks and the per-scaffold  $\text{p}K_{a,0}$  supplies a strong baseline. So the  $\text{p}K_a$  “help” is a permissive-closure and per-scaffold-prior effect, not the variance-saving of a well-specified prior: a useful caution that a “physics helps” outcome can be the encoder working through a soft closure. The absolute errors ( $\sim 1.6$ – $1.9$ ) are weak for  $\text{p}K_a$  and the DirectGNN control is not separately tuned, so we do not lean on this comparison—the robust second-domain result is the decomposition above; the trained comparison is a cautionary complement.

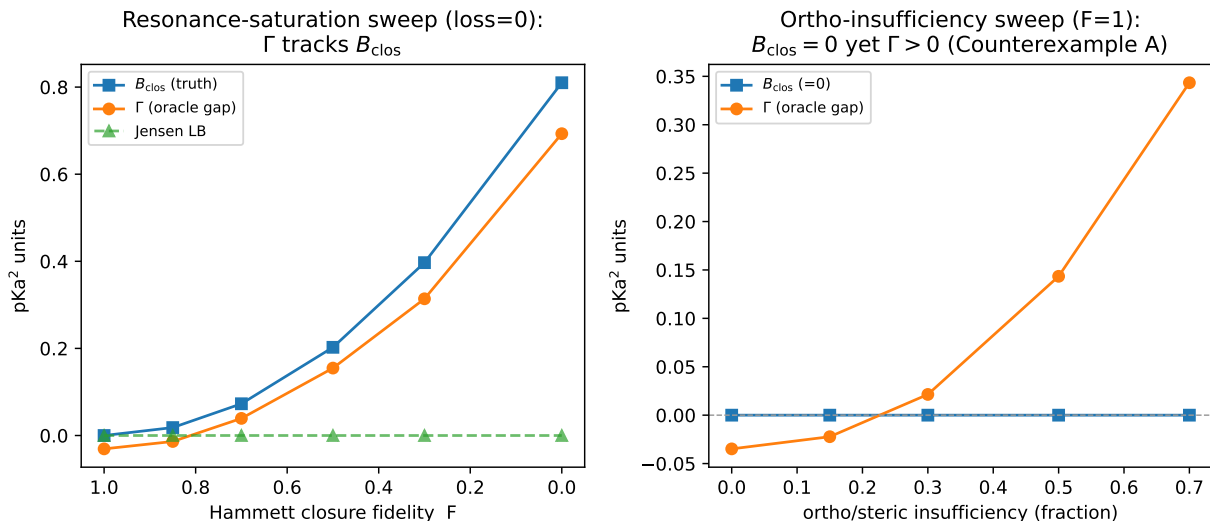


Figure 16: The diagnostic transfers to a second real closure (pKa through a Hammett LFER), on a controlled construction with known ground-truth channels. *Left*: varying closure fidelity with a sufficient intermediate, the grounding gap  $\Gamma$  tracks the closure bias  $B_{\text{closure}}$  (the assumption-light Jensen bound is flat here because the misspecification carries no mean offset). *Right*: varying an ortho/steric insufficiency channel with a *well-specified* closure,  $B_{\text{closure}} = 0$  throughout while  $\Gamma > 0$  grows—the pKa realization of the two-cause conflation of §4.3, and the reason certification rests on the decomposition, not on  $\Gamma$ .

## H Data provenance and reproducibility

**Sources.** Solubility: BigSolDB 2.0 [13]. Crystal  $T_m/\Delta H_{\text{fus}}$ : an open melting-point pool plus a group-contribution prior. True  $\sigma$ -profiles: the VT-2005 database (ingested to `sigma_profiles.csv`, 1432 compounds, with the COSMO cavity volume  $V_{\text{cosmo}}$  for the combinatorial term). These are quantum-chemically computed for a single reference conformer and protonation state; any mismatch to the experimental measurement conditions is noise in  $z^*$  that inflates  $B_{\text{insuff}}$  (an imperfect intermediate) rather than  $B_{\text{closure}}$ , so it is *conservative* for the closure-dominance verdict. Infinite-dilution activity coefficients: ThermoML. Second-domain pKa: the OPERA experimental compilation [18] (`pKa_QR.sdf`, 6502 compounds, public-domain US EPA; cited, not redistributed). The Bemis–Murcko scaffold split is produced by `prepare_data.py`; melting-point tables are auto-detected as °C or K to avoid a +273 K double-conversion.

**Compute ledger.** The headline  $\sigma$ -grounding comparison (Fig. 6, Table 6) and the data-efficiency sweep (Table 4) are full-corpus GPU runs (three seeds for the former, seed 42 for the latter); the closure/insufficiency decomposition (§6.2), the Fisher audit (§F.1), the noise floor, the encoder probe, and the temperature and mechanism diagnostics of §E are CPU-reproducible from committed artifacts. Each figure and table names the script that regenerates it from `results/`; large prediction CSVs and model checkpoints are deposited separately ([Zenodo DOI]) rather than in the repository.

**Seeds and determinism.** Controlled comparisons use seeds {42, 43, 44} and are reported on a cross-arm intersection lock (rows supervised and finite in every arm); the decomposition is seed-independent (its closure has no learnable parameters).

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